

Large Language Models in Research and Science: Foundations, Tools, and Practice

Frank Hause

Summer Semester 2026

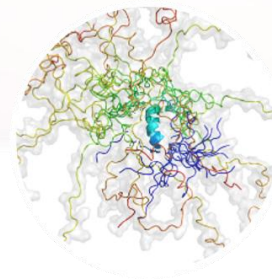


MARTIN-LUTHER-UNIVERSITÄT
HALLE-WITTENBERG

RTG 2467

Intrinsically Disordered Proteins
– Molecular Principles, Cellular
Functions, and Diseases

Speaker of the RTG: Prof. Dr. Andrea Sinz
Scientific Coordinator: Dr. Oleksandr Sorokin



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German Research Foundation

Course Outline

Part I: Foundations & Concepts

Part II: Tools & Hands-On

Part III: Limitations & Ethics

Script:



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Part I: Foundations & Concepts



Part I: Foundations and Concepts

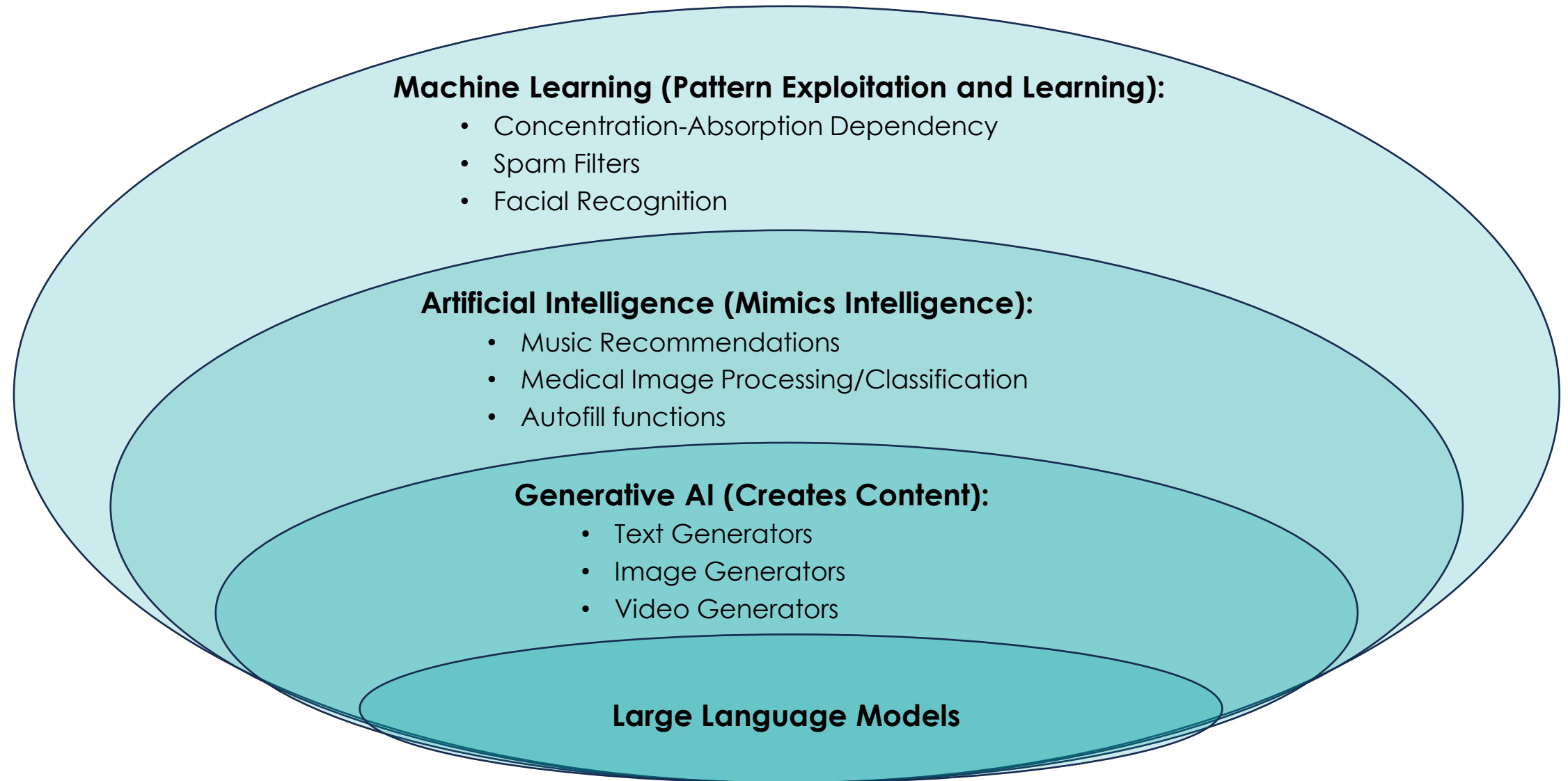
- **Concepts: Machine Learning, Artificial Intelligence, Generative AI, Large Language Models**
- **Working Principles of Large Language Models**
- **Use Cases:**
 - Text Generation
 - Knowledge Acquisition and Organization
 - Statistical Analysis and Coding
 - Agentic AIs
 - Recreation and Psychological Health
- **Prompt Engineering**
 - Prompt Anatomy (Operating Language, Clarity, Role Prompting, Constraint Setting)
 - Priming a Chat
 - Iterative Prompting and Feedback Loops
 - Prompt Evaluation and Refinement

Foundations



- Concepts of ML
- Working Principles of LLMs

Machine Learning and AI



Working Principles of LLMs: Tokenization & Embedding

Training Input:

„Artificial intelligence models learn from data to make predictions.”



Tokenized Input:

[Artificial] [intelligence] [model] [s] [learn] [from] [data] [to] [make] [prediction] [s] [.]

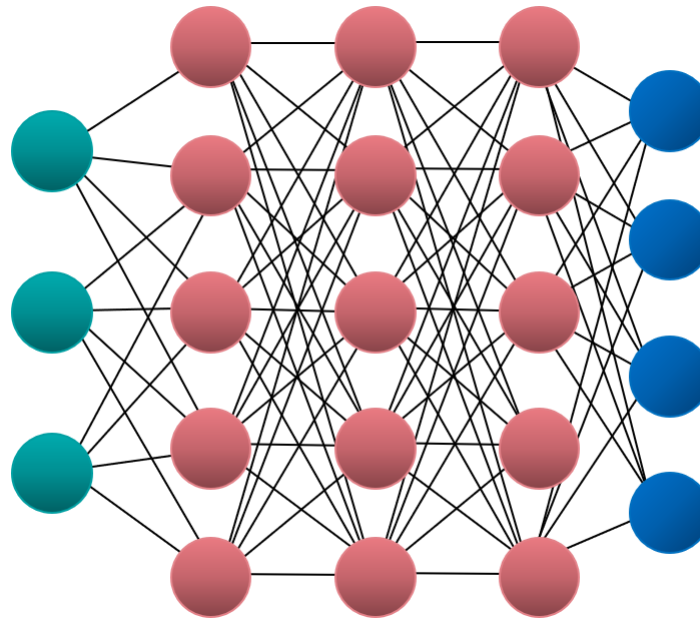


Embedding:

Token	Example Embedding
Artificial	[0.42, -0.31, 0.77, ...]
intelligence	[0.55, 0.12, 0.63, ...]
models	[0.60, -0.45, 0.22, ...]
s	[0.71, 0.05, -0.40, ...]
...	...

Working Principles of LLMs: Architecture I

Neural Network



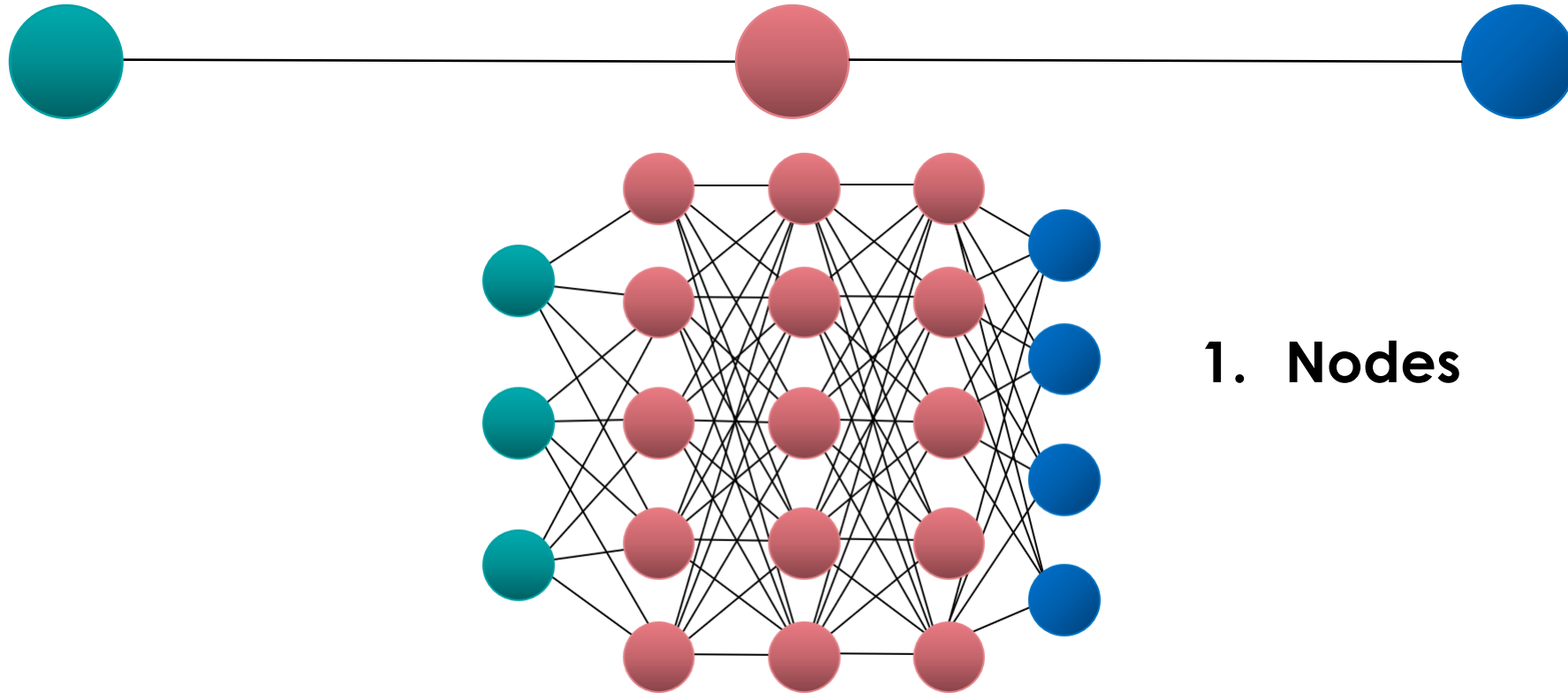
Working Principles of LLMs: Architecture II

Neural Network

Input Layer

Hidden Layer

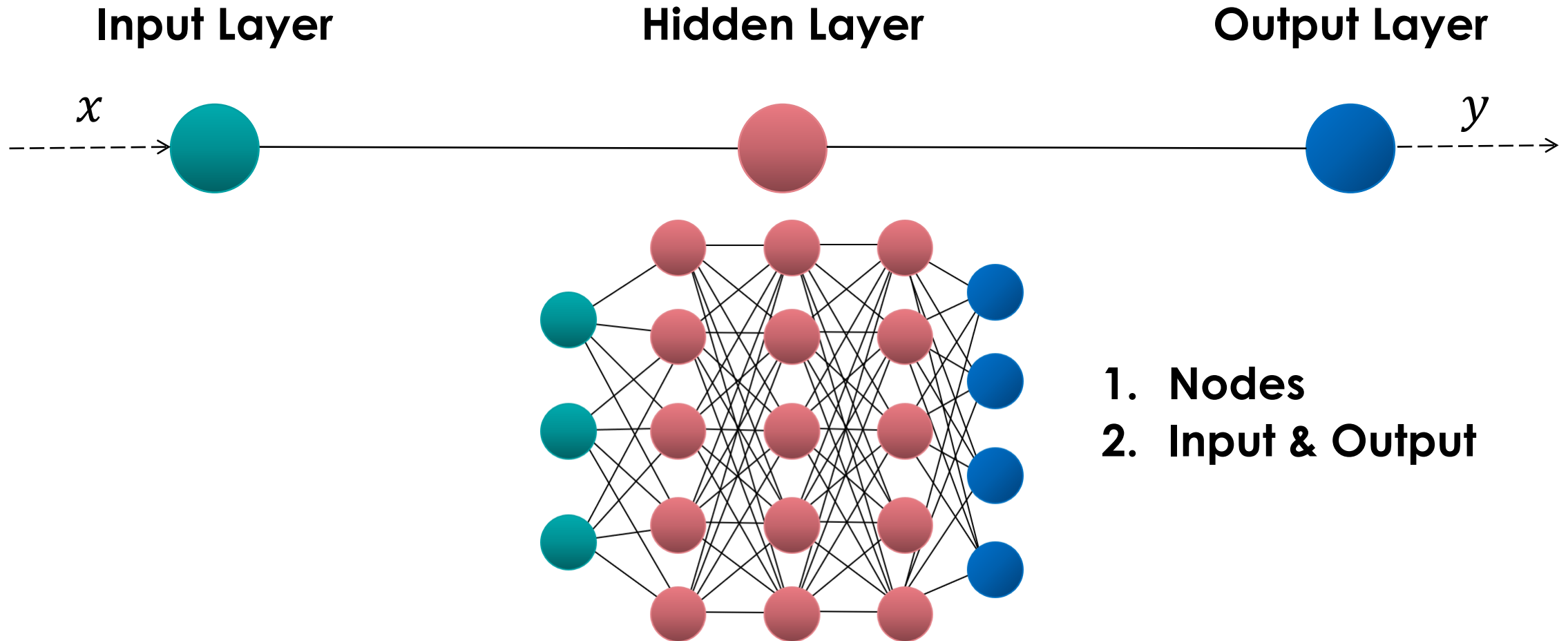
Output Layer



1. Nodes

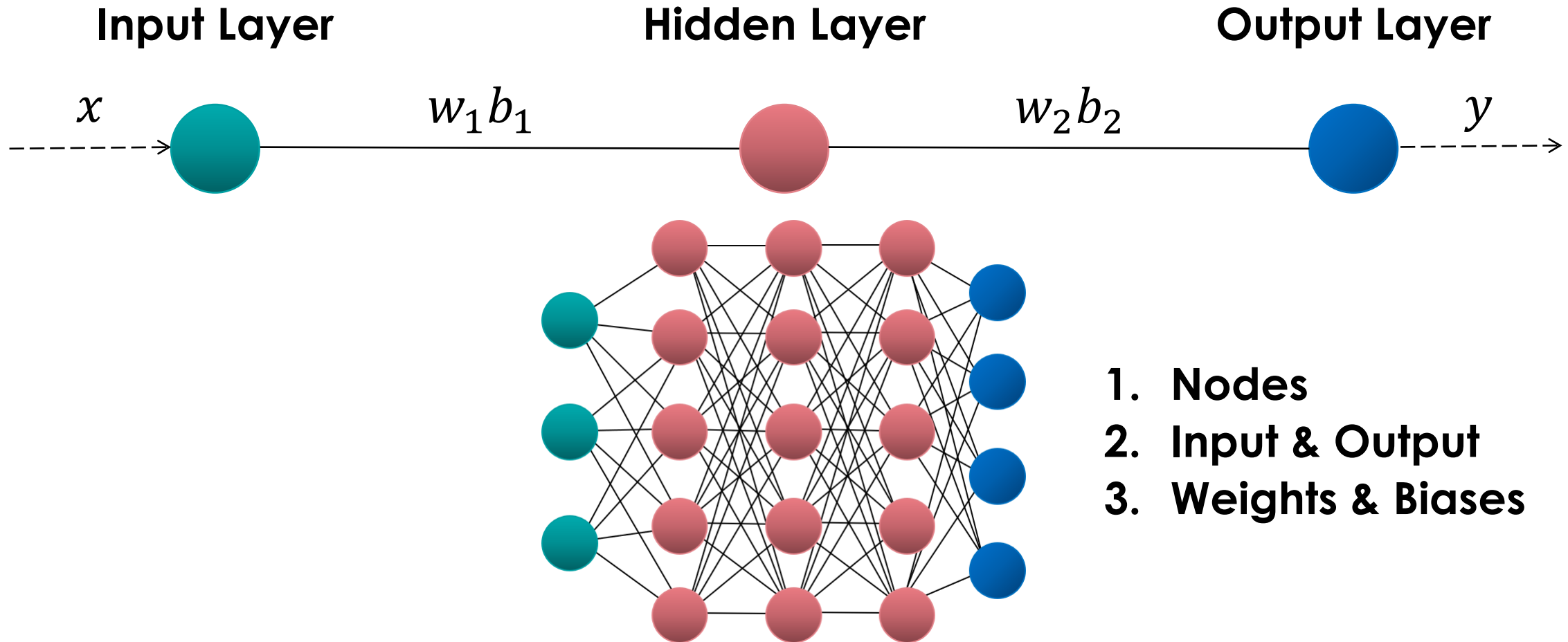
Working Principles of LLMs: Architecture II

Neural Network



Working Principles of LLMs: Architecture IV

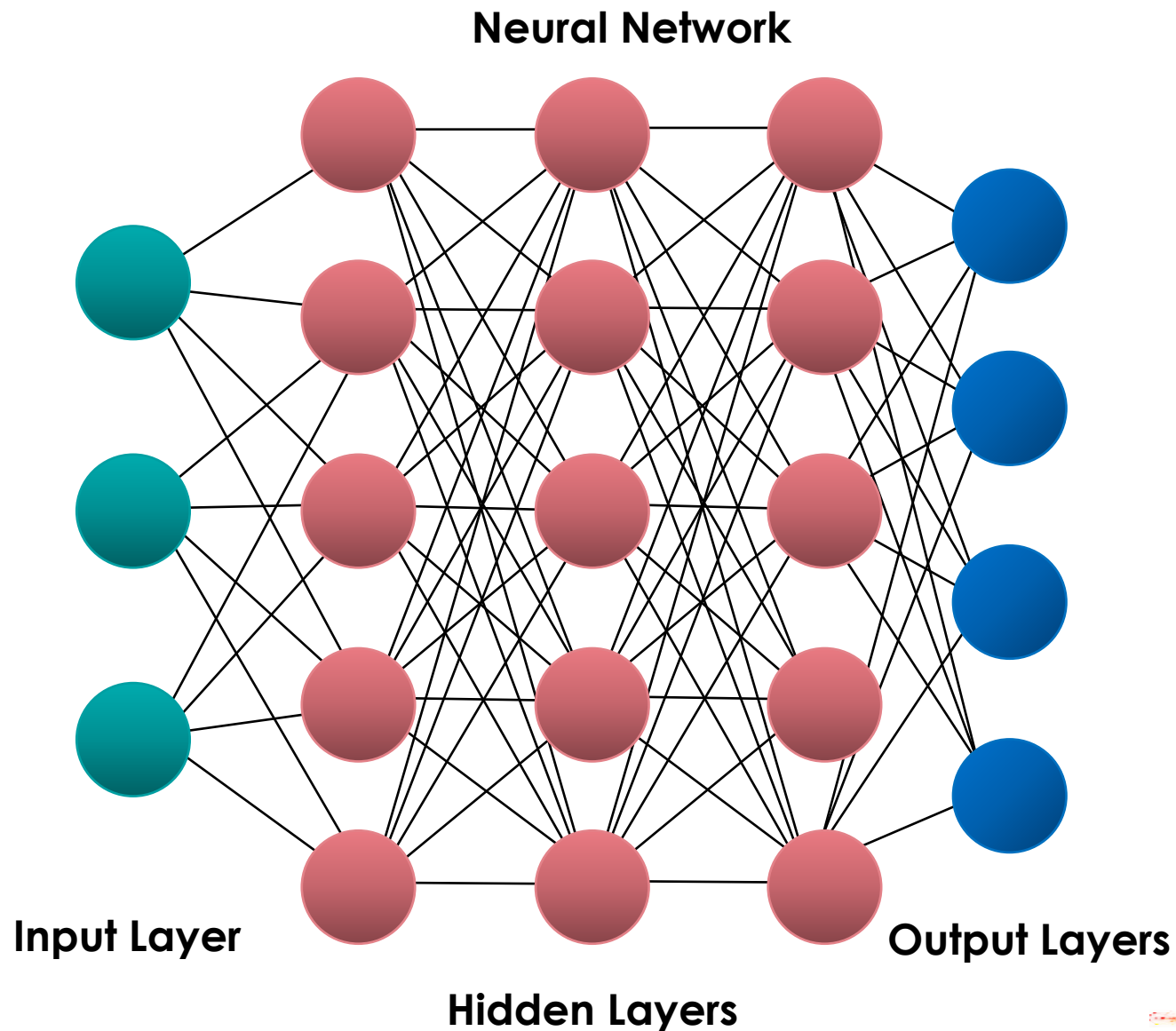
Neural Network



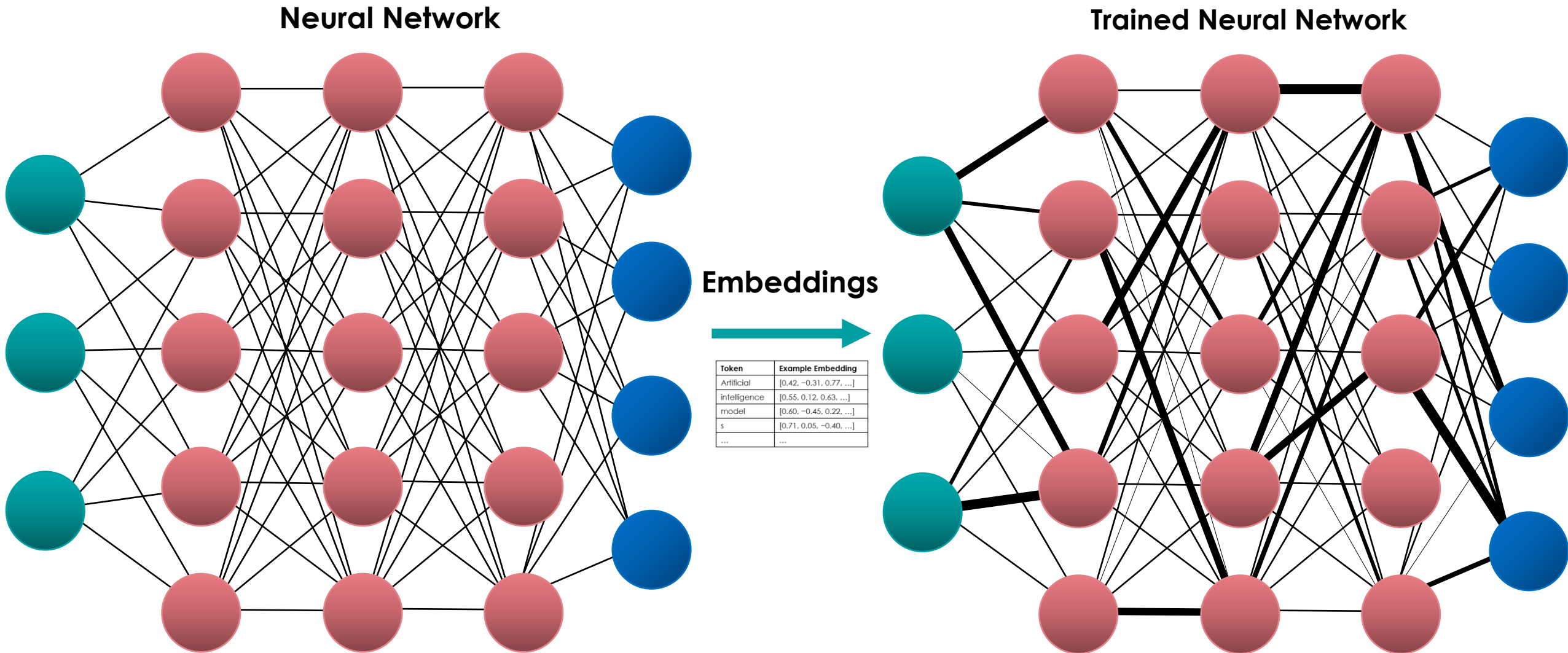
Working Principles of LLMs: Training I

Embedding:

Token	Example Embedding
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...	...

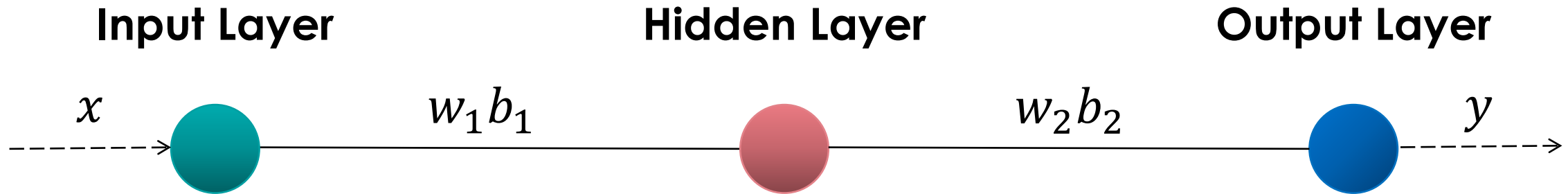


Working Principles of LLMs: Training II



Working Principles of LLMs: Training IV

Neural Network

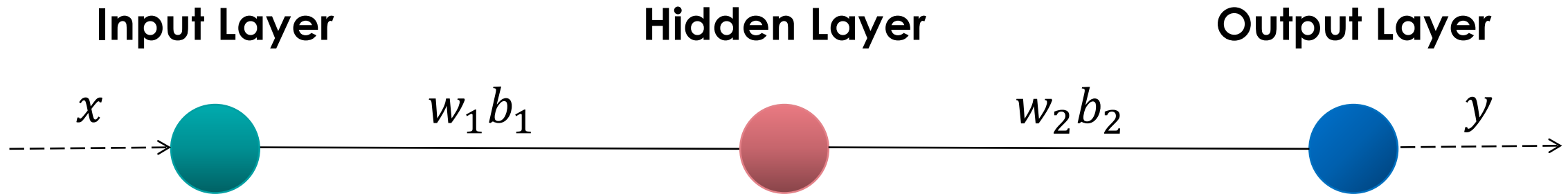


1. Pre-Activation Step

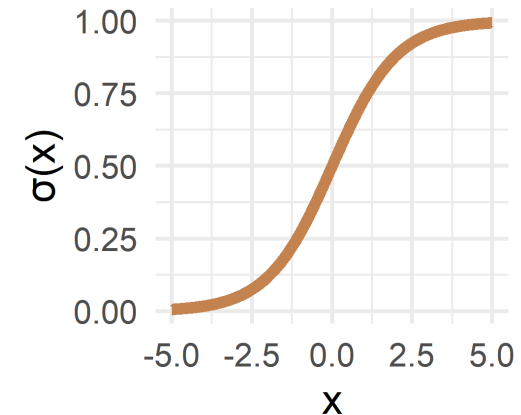
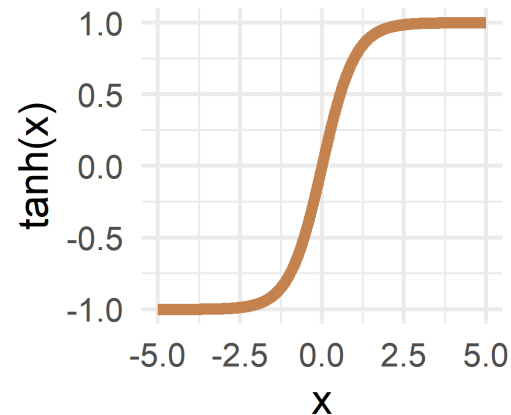
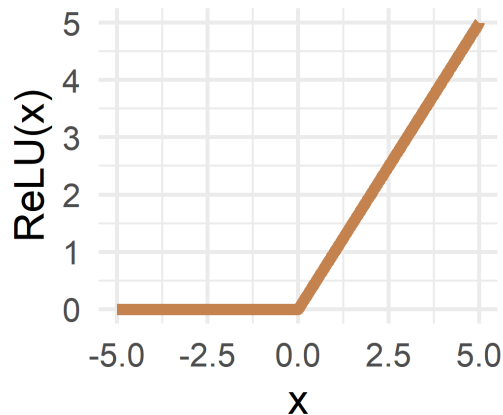
$$z = \text{weight} * \text{input} + \text{bias}$$

Working Principles of LLMs: Training III

Neural Network

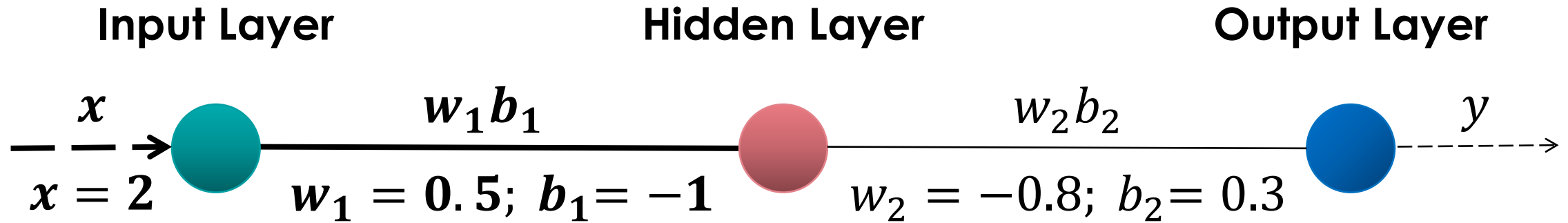


2. Activation Step $h = f_{activation}(z)$



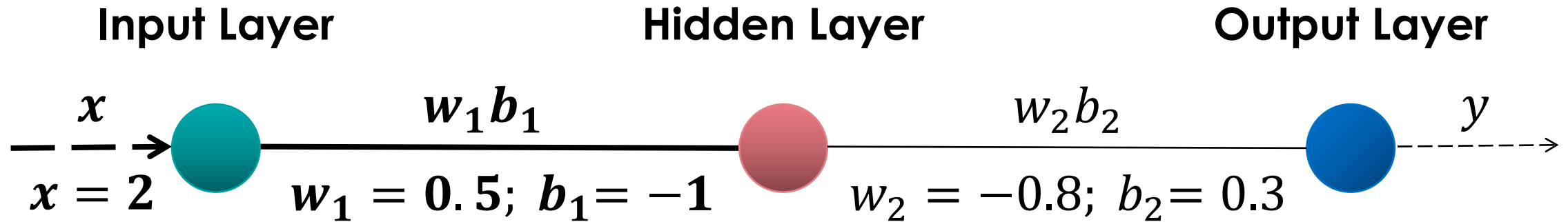
Working Principles of LLMs: Training Example I

Neural Network



Working Principles of LLMs: Training Example II

Neural Network

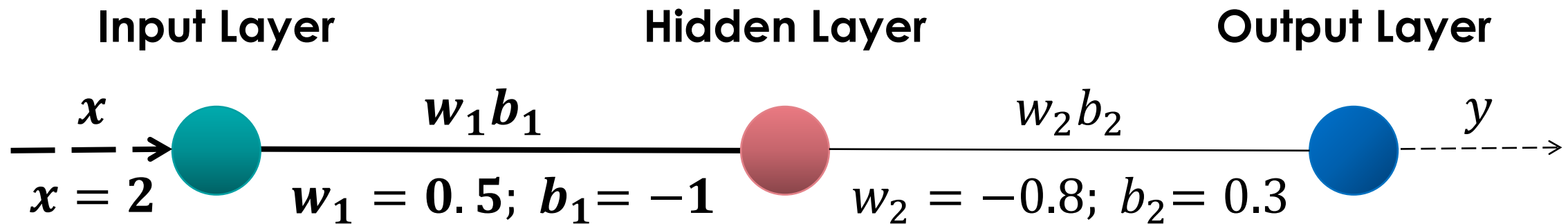


1. Pre-Activation Step

$$z_1 = w_1 * x + b_1$$

Working Principles of LLMs: Training Example III

Neural Network

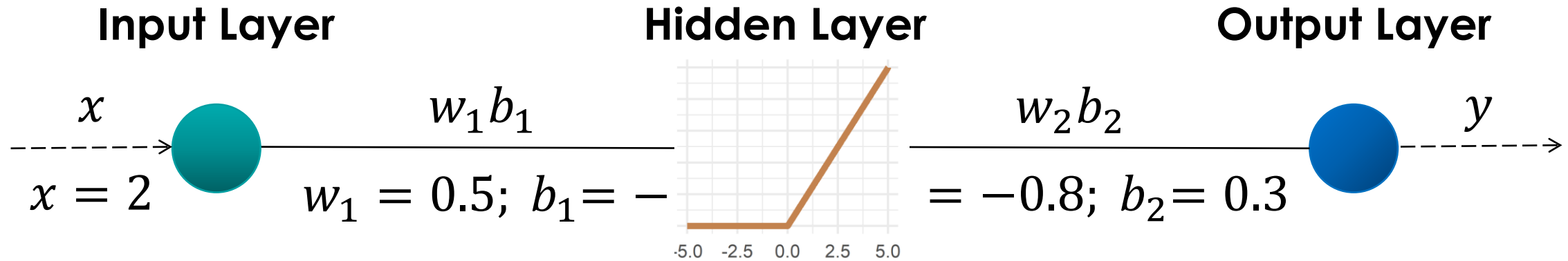


1. Pre-Activation Step

$$z_1 = w_1 * x + b_1 = 0.5 * 2 + (-1) = 0$$

Working Principles of LLMs: Training Example IV

Neural Network

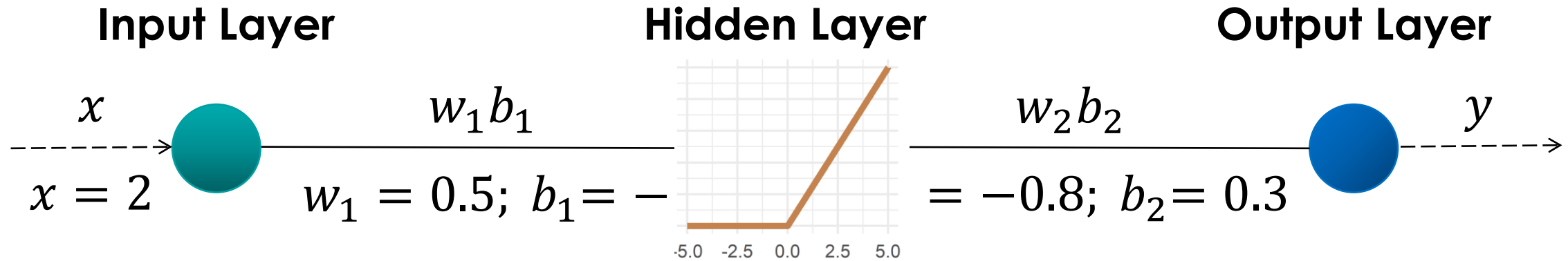


2. Activation Step

$$z_1 = 0 \quad h_1 = f_{activation}(z_1) = ReLU(z_1)$$

Working Principles of LLMs: Training Example V

Neural Network

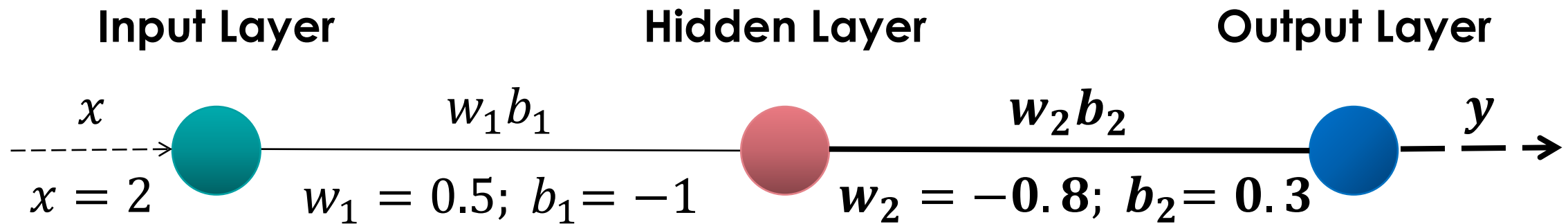


2. Activation Step

$$z_1 = 0 \quad h_1 = f_{activation}(z_1) = ReLU(z_1) = \max(0, 0) = 0$$

Working Principles of LLMs: Training Example VI

Neural Network

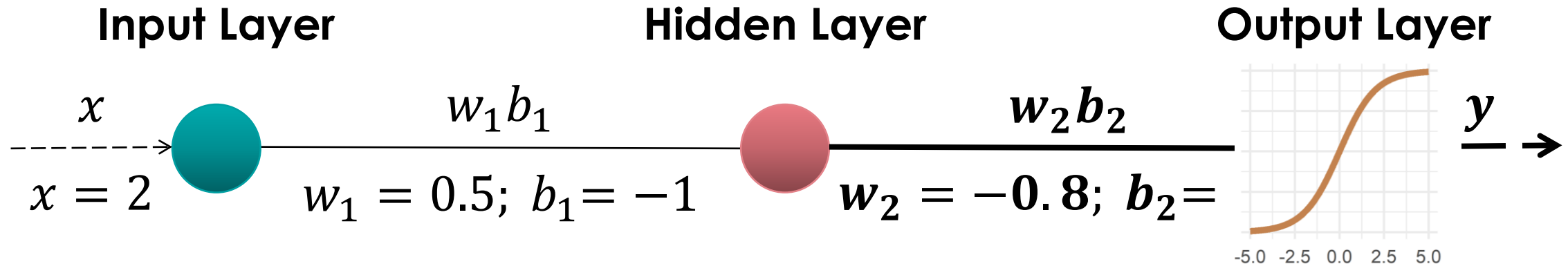


3. Pre-Activation Step

$$h_1 = 0 \quad z_2 = w_2 * h_1 + b_2 = -0.8 * 0 + 0.3 = 0.3$$

Working Principles of LLMs: Training Example VII

Neural Network

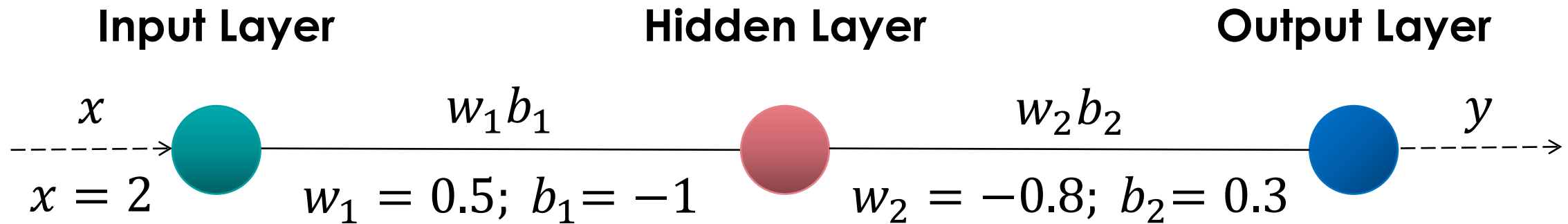


4. Activation Step

$$z_2 = 0.3 \quad y = f_{activation}(z_2) = \sigma(z_2) = \frac{1}{1 + e^{z_2}} \approx 0.574$$

Working Principles of LLMs: Training Example VIII

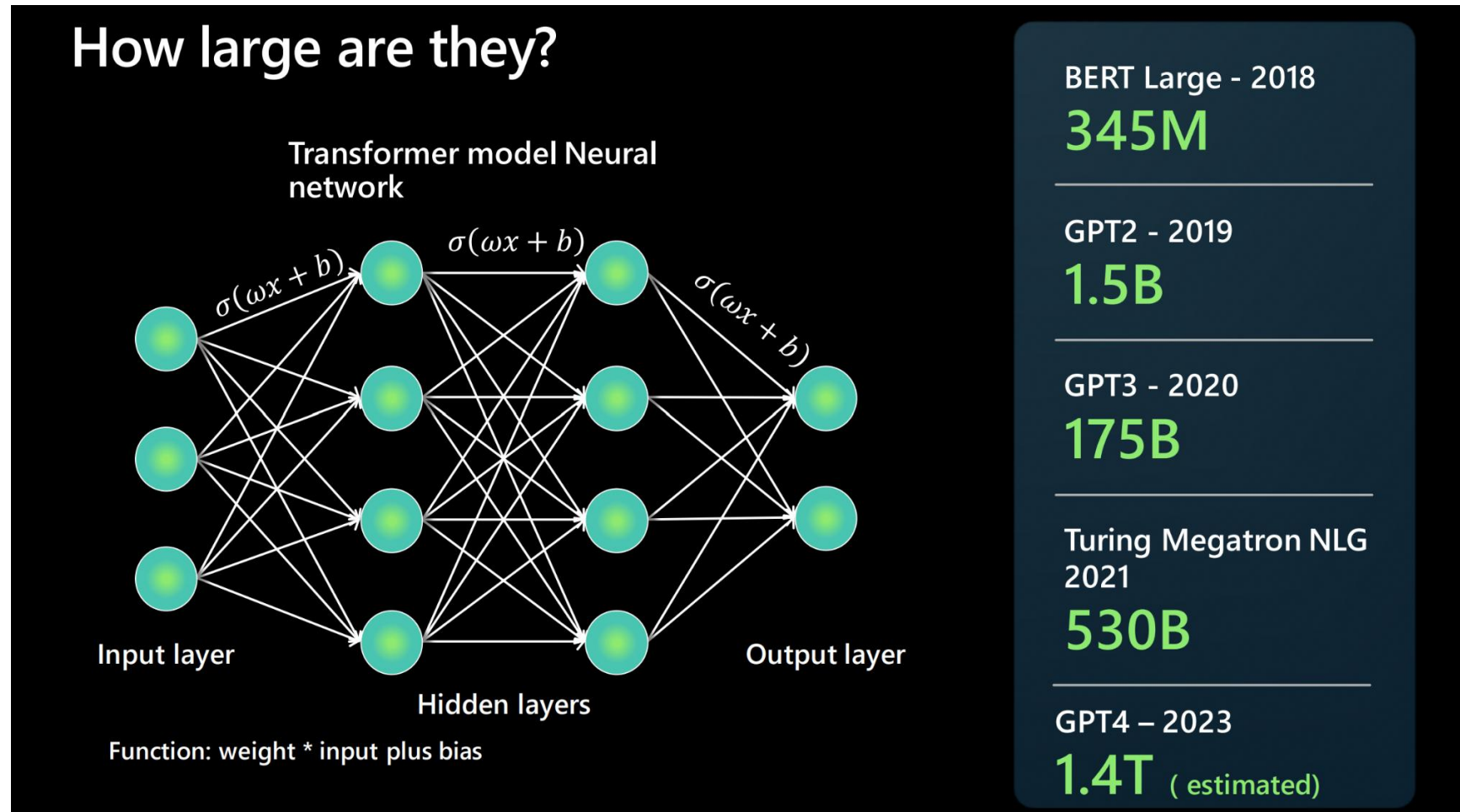
Neural Network



Additional Properties to Consider:

- Parameter Updating Rules
- Loss Gradient
- Backpropagation
- ...

Working Principles of LLMs: Size



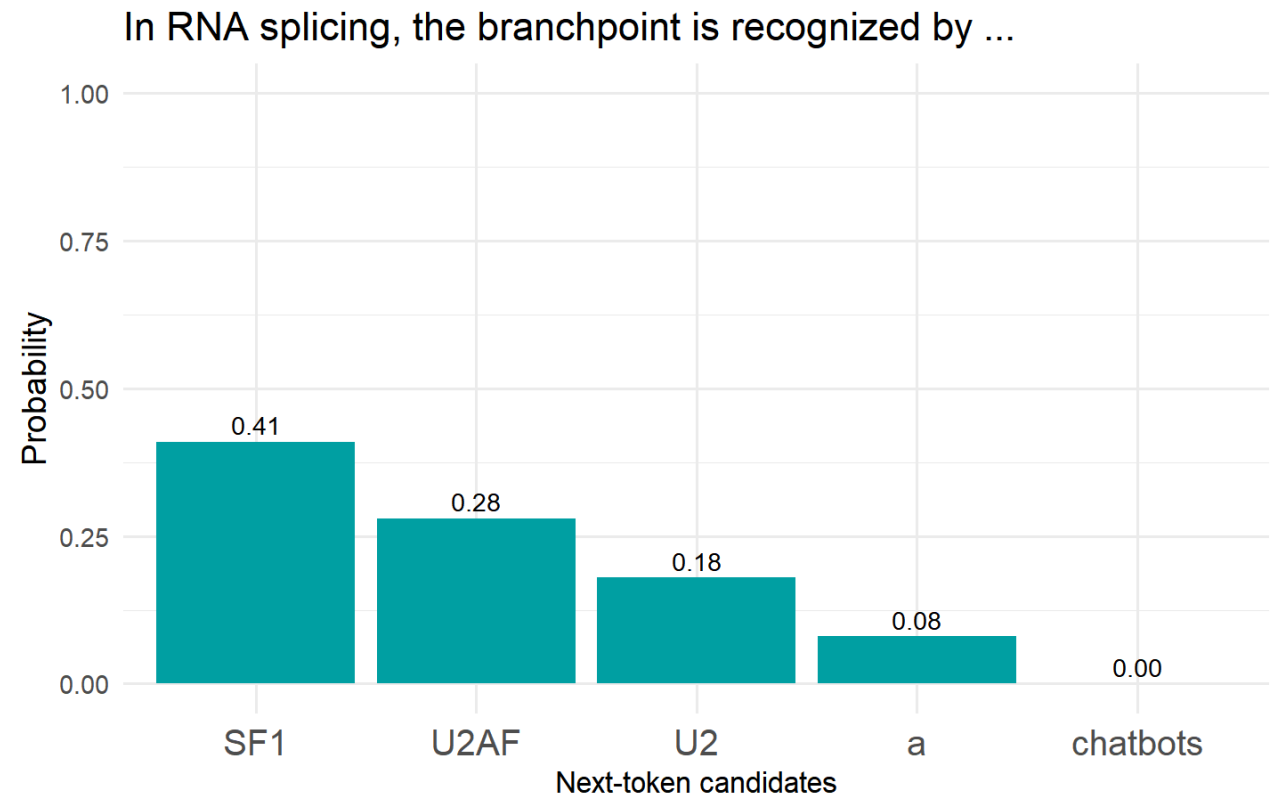
<https://microsoft.github.io/Workshop-Interact-with-OpenAI-models/llms/>

Working Principles of LLMs: Principle

What does an LLM actually do?
 $p(\text{next token} \mid \text{previous token})$

With implicit context given by:

- grammar
- domain knowledge
- logic patterns
- world structure
- argumentation
- mathematical reasoning
- emotional tone
- scientific facts
- coding conventions
- causal correlations

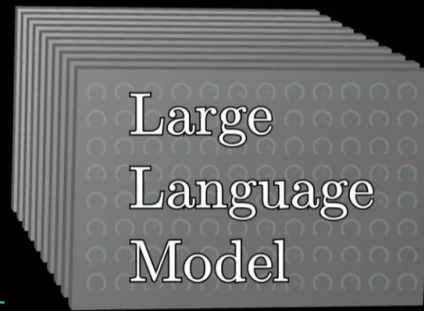


Working Principles of LLMs: Principle

What follows is a conversation between a user and a helpful, very knowledgeable AI assistant.

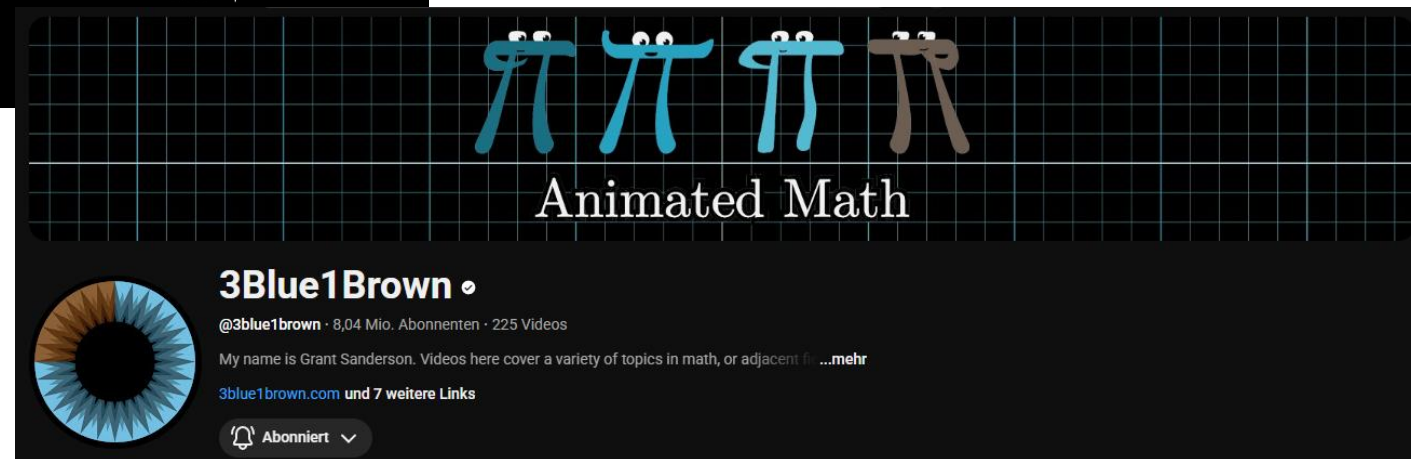
User: Give me some ideas for what to do when visiting Santiago.

AI Assistant: Sure, there are plenty of things to do in Santiago! One option could be to take a walking tour of **the** _____



the	99%
some	0%
historic	0%
its	0%
downtown	0%
Santiago	0%
historical	0%
La	0%
Cer	0%
all	0%
Old	0%

<https://www.youtube.com/watch?v=LPZh9BOjkQs>



Use Cases



- Text Generation
- Knowledge Acquisition & Organization
- Statistical Analysis & Coding
- Reflection & Psychological Health

Use Cases: Text Generation I



LLMs enable:

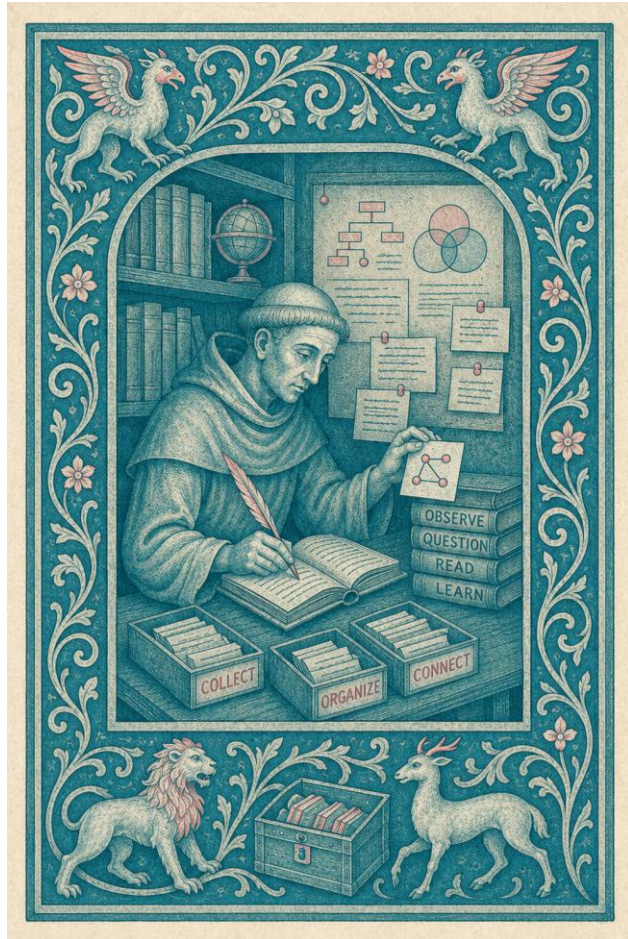
- content creation: generating natural language from prompts
- content transformation: rewriting, simplifying, enriching
- content continuation: predicting plausible next sentences or sections
- content adaptation: different tones, audiences, or domains

Use Cases: Text Generation II

Performance on different tasks

Strong	Medium	Weak
Create de novo content structure (outlines, section hierarchies, etc.)	Improve terminology consistency	Resolve inconsistencies in the line of argument
Translations	Provide alternative formulations	Enhance logical flow
Language polishing & grammar check-ups		
Improve professional communication (emails, letters)		
Adjust text to meet scientific standards		

Use Cases: Knowledge Acquisition & Organization I



LLMs enable:

- knowledge extraction: pulling key facts, concepts, and relationships from papers
- literature navigation: identifying relevant sources, themes, and research directions
- concept linking: connecting ideas across studies, fields, or disciplines
- rapid summarization: condensing long texts into focused, digestible insights
- gap detection: highlighting missing information, contradictions, or open questions

Use Cases: Knowledge Acquisition & Organization II

Performance on different tasks

Strong	Medium	Weak
identifying major topics and research directions	suggesting plausible missing links in a research topic	accurate citation retrieval (hallucinations)
extracting key facts from papers, abstracts, and long texts	highlighting knowledge gaps (but often tentatively)	verifying factual correctness without external tools
		deep source-critical evaluation
		retrieving specific details (numbers, parameters, rare facts)

Use Cases: Statistical Analysis and Coding I



LLMs enable:

- statistical guidance: explaining methods, assumptions, and choosing appropriate tests
- workflow design: outlining full analysis pipelines from raw data to results
- code generation: producing scripts in any programming language tailored to your data structure
- code transformation: rewriting, optimizing, or translating code between languages

Use Cases: Statistical Analysis and Coding II

Performance on different tasks

Strong	Medium	Weak
explaining statistical concepts (tests, distributions, assumptions)	rewriting, optimizing, and translating code across languages	performing calculations itself (language models $\neq$ math models)
designing analysis pipelines (from raw data to model output)	debugging obvious syntax errors or misused functions	performing actual statistical inference
writing clean code for common tasks (wrangling, plotting, modeling)	interpreting model outputs (esp. mixed models, survival analysis, GLMs)	guaranteeing numerical correctness in code output
	suggesting alternative modeling strategies	producing fully verified, error-free scripts for advanced ML or Bayesian workflows

Use Cases: Agentic AIs

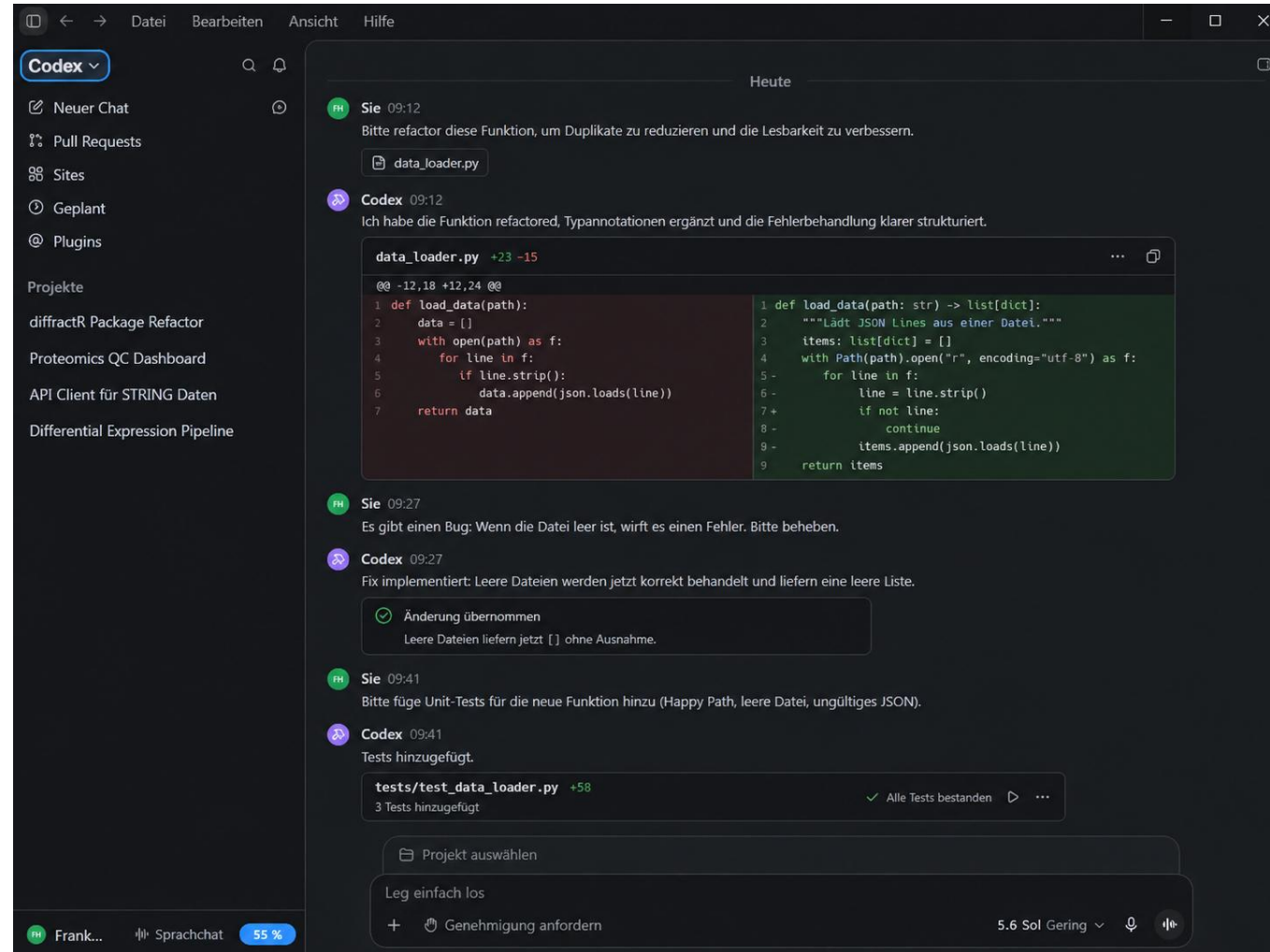


Als enable:

- goal-directed planning: decomposing complex objectives into manageable, sequential tasks
- autonomous
- tool use: interacting with software, databases, APIs, files, and web resources to execute tasks
- multi-step workflow execution: carrying out complete processes rather than generating a single response
- adaptive decision-making: evaluating intermediate results, revising the strategy, and selecting appropriate next actions
- persistent task management: tracking progress, maintaining context, and coordinating longer-running or recurring activities

Use Cases: Agentic AIs

Example: ChatGPT Codex



Use Cases: Agentic AIs

Performance on different tasks

Strong	Medium	Weak
Almost independent fulfillment of any given analytical or programming task		

Use Cases: Agentic AIs

Performance on different tasks

Strong	Medium	Weak
Almost independent fulfillment of any given analytical or programming task		

Problem:

Agentic AIs are often only to be controlled by the plausibility of the results → Explanatory Gap

Use Cases: Reflection and Psychological Health I



LLMs enable:

- immediate, low-threshold emotional reflection when no one else is reachable
- externalizing overwhelming thoughts, helping reduce rumination
- structured prompts and calm dialogue that bring order into acute distress
- a sense of “somebody being around”

Use Cases: Recreation and Psychological Health

PhD students are at high risk for psychiatric health problems



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Research Policy

journal homepage: www.elsevier.com/locate/respol



Work organization and mental health problems in PhD students

Katia Levecque^{a,b,*}, Frederik Anseel^{a,b,c}, Alain De Beuckelaer^{d,e,a},
Johan Van der Heyden^{f,g}, Lydia Gisle^f

A B S T R A C T

Research policy observers are increasingly concerned about the potential impact of current academic working conditions on mental health, particularly in PhD students. The aim of the current study is three-fold. First, we assess the prevalence of mental health problems in a representative sample of PhD students in Flanders, Belgium ($N = 3659$). Second, we compare PhD students to three other samples: (1) highly educated in the general population ($N = 769$); (2) highly educated employees ($N = 592$); and (3) higher education students ($N = 333$). Third, we assess those organizational factors relating to the role of PhD students that predict mental health status. Results based on 12 mental health symptoms (GHQ-12) showed that 32% of PhD students are at risk of having or developing a common psychiatric disorder, especially depression. This estimate was significantly higher than those obtained in the comparison groups. Organizational policies were significantly associated with the prevalence of mental health problems. Especially work-family interface, job demands and job control, the supervisor's leadership style, team decision-making culture, and perception of a career outside academia are linked to mental health problems.

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RESEARCH ARTICLE

"How is your thesis going?"–Ph.D. students' perspectives on mental health and stress in academia

Julian Friedrich^{1,2,*}, Anna Bareis^{1,2}, Moritz Bross^{1,2}, Zoé Bürger^{1,2}, Álvaro Cortés Rodríguez^{1,2}, Nina Effenberger^{1,2}, Markus Kleinhansl^{1,2}, Fabienne Kremer^{1,2}, Cornelius Schröder^{1,2}

General health and stressors

The mean of the sum score for PHQ-2 in our study was 2.32 which is below the cut-off of three for major depression [26]. Yet, 33.1% of the participants were above the cut-off. For the GAD-7, the sum score for the study's sample was 8.49. Cut points of 5 might be interpreted as mild, cut points of 10 as moderate and 15 as severe levels of anxiety [27], which implies a mild risk level for generalized anxiety with the suggestion of a follow-up examination in this sample.

When asking for mental disorders, we found that 19.9% of the participants ($n = 99$) have already been diagnosed with a mental disorder and 15.5% ($n = 77$) are currently in psychotherapy. The sum score for the Perceived Stress Scale (PSS) of 7.79 (with $Min = 0$, $Max = 15$) was

Use Cases: Recreation and Psychological Health

Use of LLMs because of psychological distress is very common

ChatGPT sees one million users in mental distress each week, APS in The Daily Aus

Artificial Intelligence (AI) | Suicide prevention | Youth mental health

<https://psychology.org.au/insights/chatgpt-sees-one-million-users-in-mental-distress>

ChatGPT: More than a million users show signs of mental health distress and mania each week, internal data suggest

Adrian O'Dowd

<https://www.bmj.com/content/bmj/391/bmj.r2290.full.pdf>

Over one in three using AI Chatbots for mental health support, as charity calls for urgent safeguards

Posted 18th November 2025

Artificial intelligence is already being widely consulted for mental health support in the UK, with more than one in three adults (37%) [1] saying they've used an AI chatbot to support their mental health or wellbeing. The surprising rapid pace of adoption has prompted experts to call for safeguards to ensure people receive accurate, safe and appropriate information.

<https://mentalhealth-uk.org/blog/over-one-in-three-using-ai-chatbots-for-mental-health-support-as-charity-calls-for-urgent-safeguards/>

Use Cases: Recreation and Psychological Health

Current LLMs are not safe for handling acute psychological crises

Between Help and Harm: An Evaluation of Mental Health Crisis Handling by LLMs

ADRIAN ARNAIZ-RODRIGUEZ, ELLIS Alicante, Spain

MIGUEL BAIDAL, ELLIS Alicante, Spain

ERIK DERNER*, ELLIS Alicante, Spain

JENN LAYTON ANNABLE, School of H

MARK BALL, School of Psychology, Public

MARK INCE, Mental Health Practitioner, S

ELVIRA PEREZ VALLEJOS, School of C
United Kingdom

NURIA OLIVER, ELLIS Alicante, Spain

designed protocol for assessing response appropriateness. We systematically benchmark three state-of-the-art LLMs (gpt-4o-mini, gpt-5-nano, and llama-4-scout) for their ability to classify crisis types and generate safe, appropriate responses. Our evaluation pipeline assesses both detection and response quality using a 5-point Likert scale ranging from harmful to fully appropriate. The results reveal that while LLMs are highly consistent and generally reliable in addressing explicit crisis disclosures, significant risks remain. A non-negligible proportion of responses, especially in self-harm and suicidal ideation categories, are rated as inappropriate or harmful, with responses generated by an open-weight model exhibiting higher failure rates than those generated by the commercial ones. We also identify systemic weaknesses in handling indirect or ambiguous risk signals, a reliance on formulaic and inauthentic default replies, and frequent misalignment with user context. These findings underscore the urgent need for enhanced safeguards, improved crisis detection, and context-aware interventions in LLM deployments. Our taxonomy, datasets, and evaluation framework lay the

Use Cases: Recreation and Psychological Health

Help Options during Acute Psychological Crises

Emergency Services:

110/112 (acute danger)

24/7 Crisis Hotlines:

0800-1110111 or 0800-1110222 or 116123
(Telefonseelsorge)

Outpatient and Psychiatric Services:

0345-5573611 (Universitätsklinikum Halle
— Psychiatric Crisis Care)

24/7 Walk-in (Universitätsklinikum Halle):

Julius-Kühn-Straße 7, 06112 Halle
Ernst-Grube-Straße 40, 06120 Halle

Counseling Centers:

City Halle (<https://halle.de/leben-in-halle/gesundheit/psychosoziale-hilfe>)

Studentenwerk (<https://studentenwerk-halle.de/beratung-soziales/psychosoziale-beratung/beratungsangebote>)

Prompt Engineering



- Language Awareness
- Clarity
- Role Prompting
- Constraint Setting
- Chat Priming
- Iterative Prompting
- Prompt Evaluation & Refinement
- Meta-Prompting

Prompt Engineering: Single Prompt Properties

1. Language Awareness

- **Output Language:**
 - Prompt in the language you want the result in
 - Mixed-language prompts will degrade output quality
 - Consider general language settings (NotebookLM)
- **Training Bias Awareness:**
 - Models are typically strongest in English for technical and scientific tasks (e.g. coding, math, STEM writing ...)
 - Narrative, cultural, or administrative tasks may work well in local languages (professional communication, presentations, ...)

→ Prompting Language should be chosen strategically

Prompt Engineering: Single Prompt Properties

2. Clarity

- **Avoid underspecified phrases:**
 - “Analyze the attached data.”
 - “Improve the following text.”
 - “Explain this paper.”
- **Be concise, contextualize, state motivation, set goal:**
 - “Make suggestions for analyzing the attached data with respect to identifying significant differences (assume normality) when comparing samples from condition 1 and condition 2.”
 - “Improve the following text with respect to scientific tone, comprehensibility, and precise phrasing. It is intended as part of a presentation to a scientific audience.”
 - “Explain the conclusions presented in the paper with respect to the methodological limitations of the experimental approaches.”

Prompt Engineering: Single Prompt Properties

3. Constraint Setting

- **What are constraints?**
 - Explicit limits that restrict how the model may respond
 - Define boundaries for content, style, structure, and behavior
- **Format Constraints:**
 - “Use bullet points only.”
- **Length Constraints:**
 - “Shorten this text to 150 words.”
- **Style Constraints:**
 - “Maximize comprehensibility of these bioinformatic analysis to biochemists.”

Prompting Strategies

1. Role Prompting

- **What is role prompting?**
 - Explicitly assigning an expert role or perspective to the model
 - Constrains knowledge, vocabulary, and reasoning style
- **Specify concrete roles:**
 - “Act as a bioinformatician analyzing RNA-seq data.”
 - “Act as a journal reviewer for a high-impact structural biology journal.”
 - “Act as a PhD supervisor providing critical feedback.”

Prompting Strategies

2. Chat Priming

- **What is chat priming?**
 - Providing background context before issuing the main task
 - Establishes shared assumptions, definitions, and goals
 - Especially for complex tasks
- **What to prime?**
 - Task context and objectives ("In this conversation, we aim to develop ...")
 - Domain assumptions and definitions ("Assume all data are normally distributed.")
 - Data structure, notation, or terminology (e.g. attach your tabular data)
 - Constraints that persist across turns ("Throughout this chat, keep all explanations suitable for a scientific audience.")

Prompting Strategies

3. Iterative Prompting

- **What is iterative prompting?**
 - Refining prompts based on intermediate outputs
 - Treating the model as an interactive collaborator, not a one-shot tool
- **Why this matters?**
 - Complex problems benefit from stepwise refinement
 - Early outputs reveal ambiguities or missing constraints
 - Improves precision, depth, and alignment with user intent
- **Typical iteration patterns:**
 - Correction/Refinement: "Rewrite this section with more ..."
 - Expansion: "Now add ..."
 - Restriction/Clarification: "Focus only on ..."

Prompting Strategies

3. Iterative Prompting

- **CAVE! Refrain from:**
 - Changing goals without restating constraints
 - Accumulating contradictions across turns
 - Assuming the model remembers implicit intent

Prompting Strategies

4. Prompt Evaluation and Refinement

- **Start from intent:**
 - Users often have a clear idea of what they want
 - Productivity depends on how well this intent is externalized
 - State the goal, motivation, and expected outcome explicitly
- **Do not rely on the model to infer intent:**
 - Implicit assumptions hinder progress
 - Unstated expectations lead to misaligned outputs
 - What seems “obvious” to the user is invisible to the model
- **Make assumptions explicit:**
 - Background knowledge
 - Methodological preferences
 - Scope and exclusions

Prompting Strategies

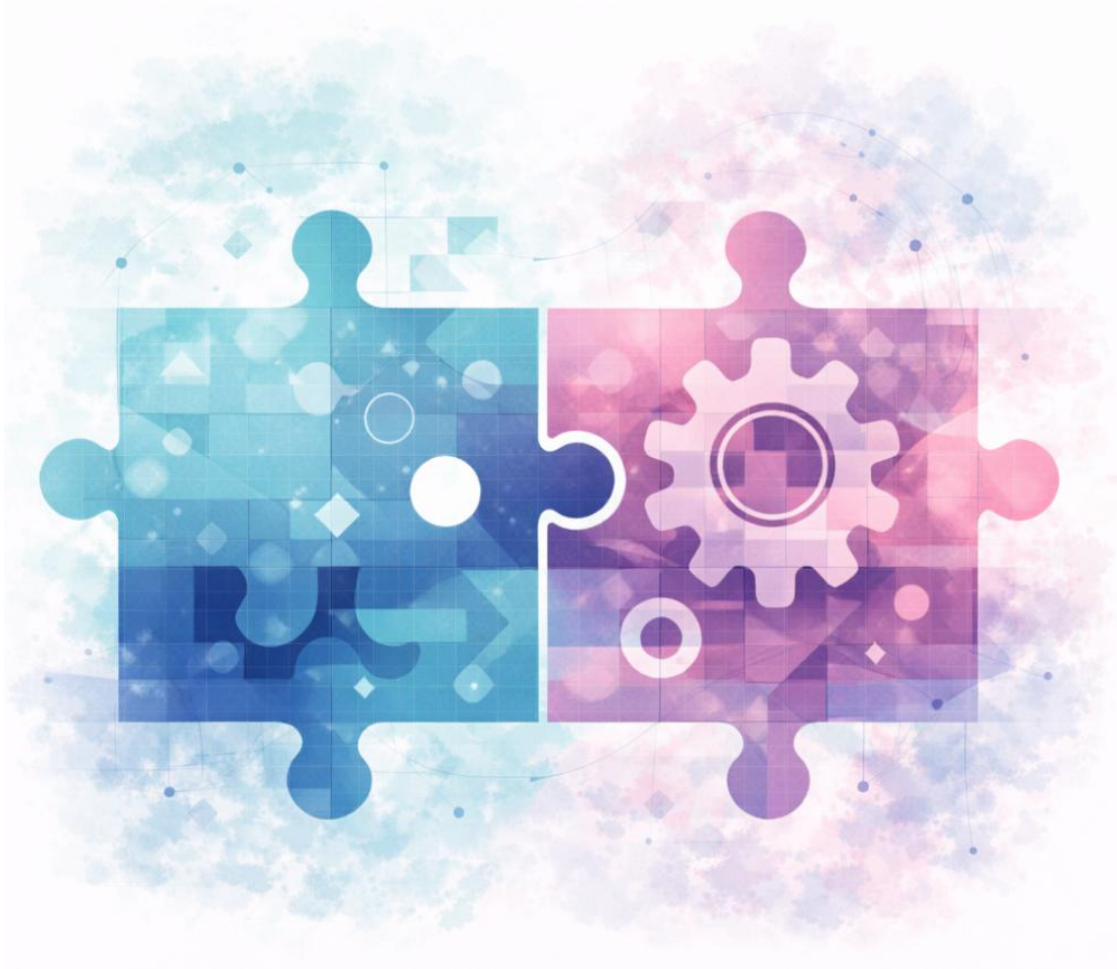
5. Meta-Prompting

- **Use one LLM to prompt another:**
 - Different models have different strengths
 - A general-purpose LLM can translate your intent into a precise, task-specific prompt
 - The resulting prompt is then passed to a specialized model or tool
- **Exploit complementary capabilities:**
 - ChatGPT: structuring, refining, and specifying complex requests
 - Perplexity: literature search and source-grounded research
 - Coding models: implementation and debugging
 - Domain-specific systems: specialized analysis
- **Let the first model optimize the prompt:**
 - Specify the goal, scope, constraints, and desired output
 - Ask it to identify ambiguities or missing information
 - Request a prompt tailored specifically to the target model
- **Example workflow:**
 - “Create a prompt for Perplexity to systematically investigate whether degree-preserving network null models retain higher-order topology in biological interaction networks.”
 - Review and refine the generated prompt
 - Submit it to Perplexity for the actual literature research



Part II: Tools & Hands-On

Part II-1: Tools

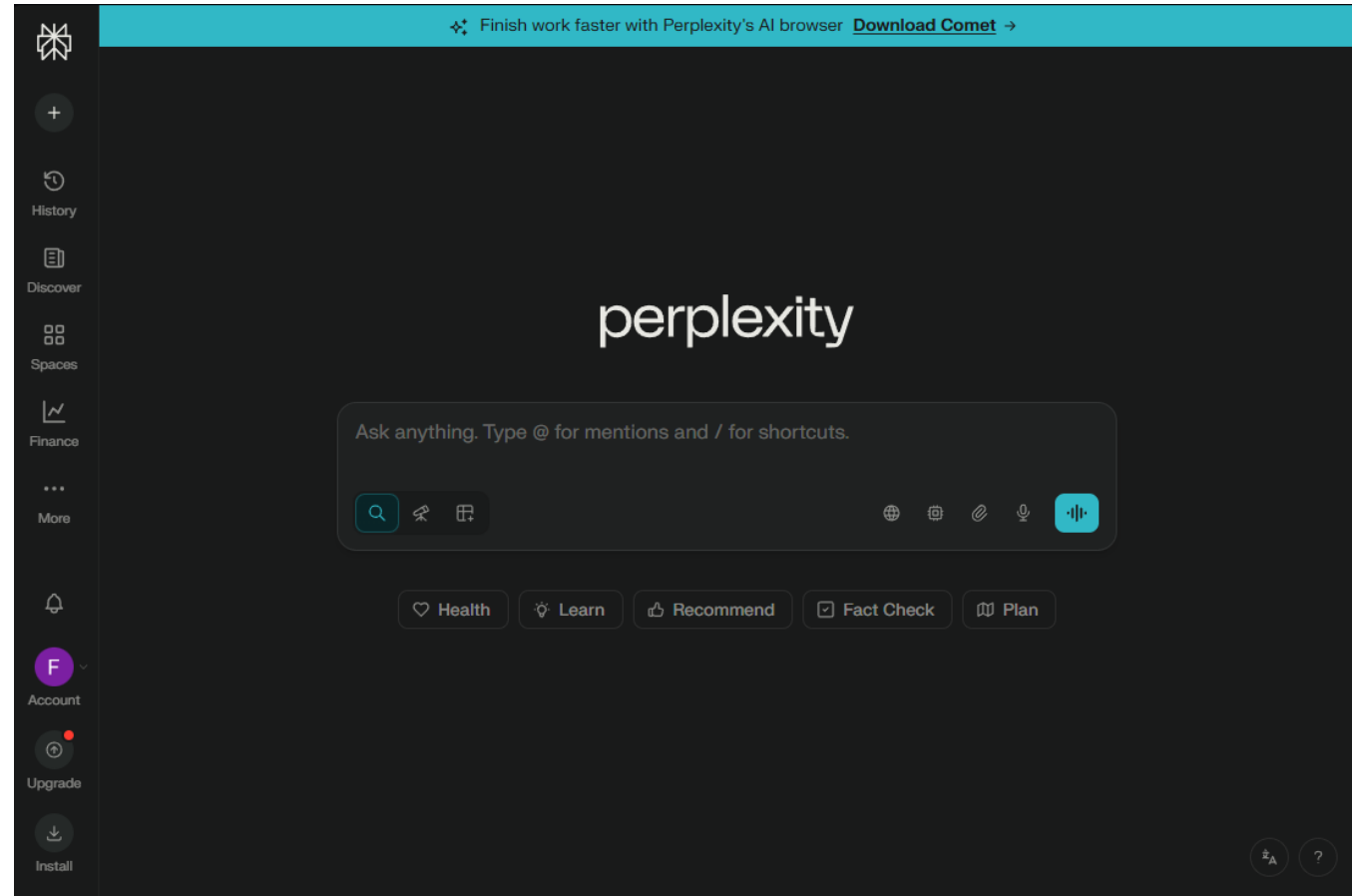


- Knowledge Mining (Perplexity.ai)
- Knowledge Transformation (ChatGPT)
- Knowledge Organization (NotebookLM)

Tools: General Considerations

- **Commercial nature of LLM providers:** LLMs are developed and operated by private companies, not public or welfare institutions.
- **Use of interactions in free services:** Interactions with free LLM services will be logged and used for model improvement or training.
- **Handling of sensitive information:** Confidential, unpublished, personal, or regulated data should not be shared with LLMs.
- **Rapid evolution of features and availability:** Tools, interfaces, and capabilities demonstrated today may change, degrade, disappear, or move behind paywalls without notice.
- **Non-stationary model performance:** Model behavior and performance can change over time due to updates, retraining, or policy adjustments; quality may improve but can also degrade or shift unpredictably.

Tools I: Perplexity.ai



perplexity.ai

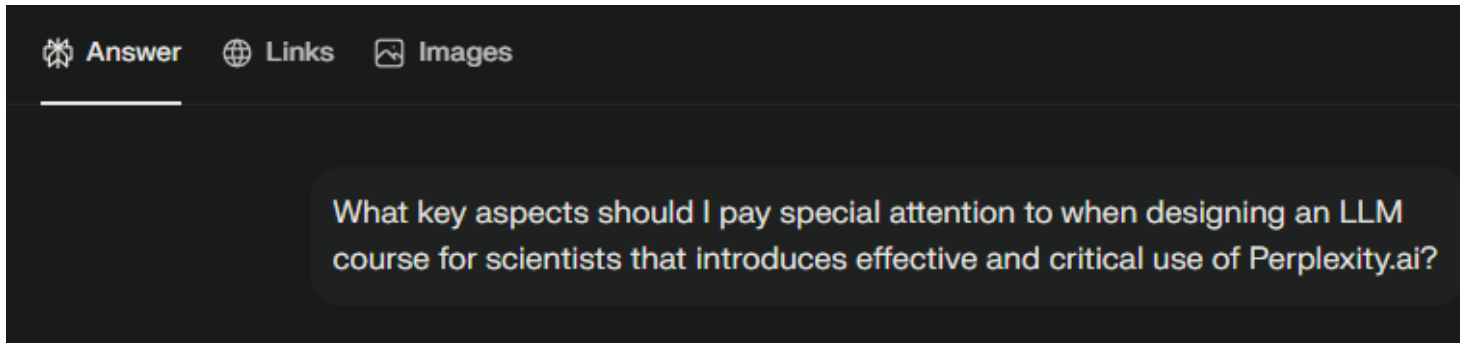
- Perplexity AI is an **AI-powered search engine** launched in 2022 (Perplexity AI, Inc., San Francisco, US)
- Processes user queries using LLMs (mostly from OpenAI, Google, and Anthropic)
- Answers are provided as **continuous text, enriched with explicit source citations**
- the platform had approximately 10 million regular users (Jan. 2024)

1. Retrieval Orientation

- **Main function**
 - User asks question → Perplexity provides a **synthesized answer with explicit references**
 - Comparison:
 - Google: Provides links **without textual synthesis**
 - ChatGPT: Provides synthesized claims **without explicit sources**
 - Perplexity: Provides synthesized claims **with traceable sources**
- **Key implication:**

Users can immediately verify the factual basis of Perplexity's claims – **at least in theory**

Tools I: Perplexity.ai



 Answer  Links  Images

What key aspects should I pay special attention to when designing an LLM course for scientists that introduces effective and critical use of Perplexity.ai?

2 steps completed >

Key Aspects for Designing an LLM Course for Scientists Using Perplexity.ai

When designing an LLM course for scientists that introduces effective and critical use of Perplexity.ai, you should focus on both the technical capabilities and the critical evaluation frameworks necessary for responsible scientific use. [socialsciencespace](#)

Understanding Core Capabilities and Limitations

Scientists need to understand Perplexity's distinctive features as an AI-powered search tool. Perplexity combines large language models with real-time web search and provides citation-grounded responses, distinguishing it from standalone LLMs like ChatGPT. Key capabilities to cover include the Sonar model's enhanced reasoning and expanded citation coverage, which places it alongside Google Gemini 2.5 Pro in web-augmented benchmarks. The platform's Comet Assistant now maintains context when users navigate between sources, streamlining research workflows. [datastudios](#) +1

Tools I: Perplexity.ai

Sources

 Answer  Links  Images

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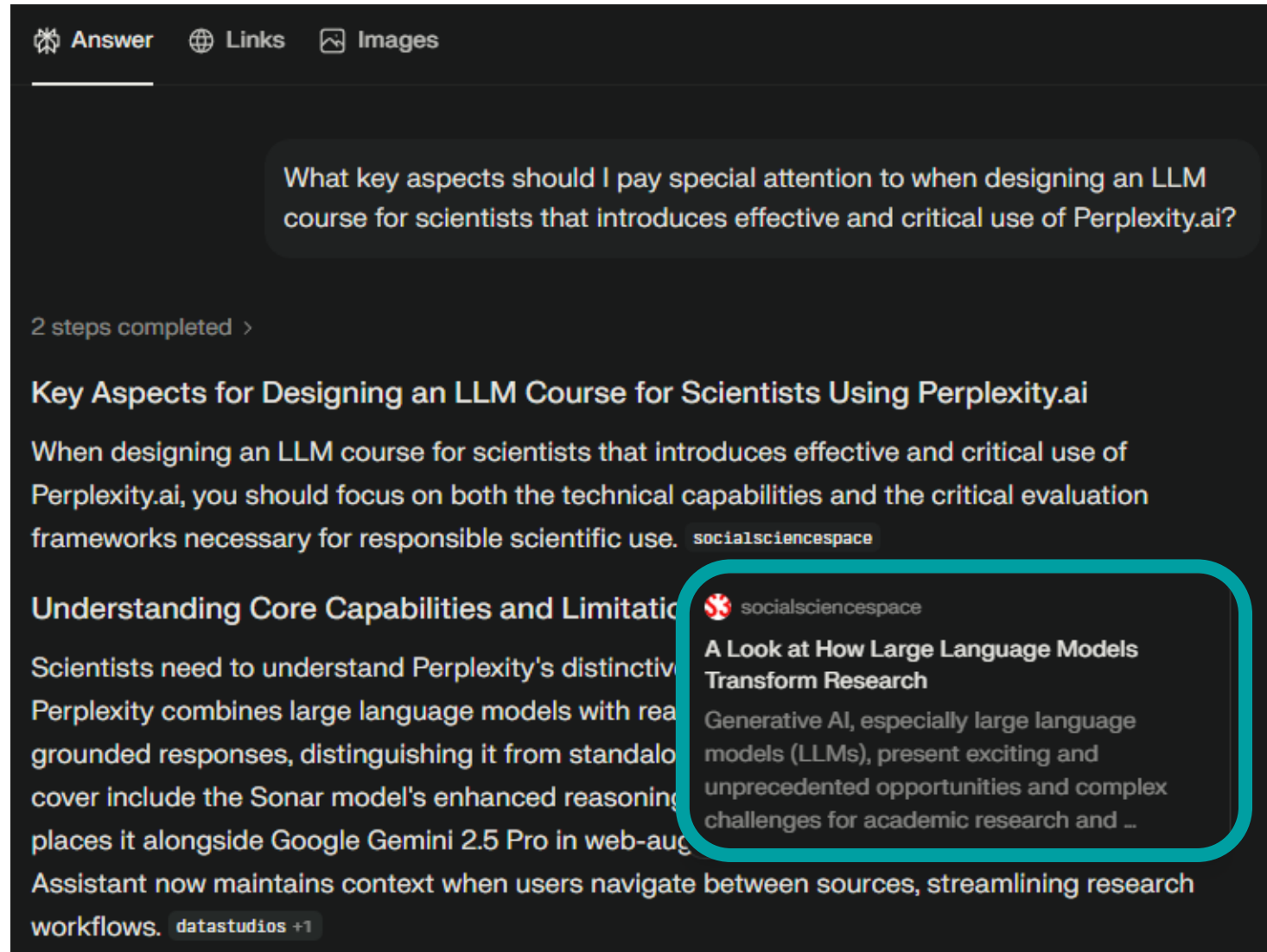
Understanding Core Capabilities and Limitations

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Synthesized Claims

Tools I: Perplexity.ai

Sources



The screenshot shows the Perplexity.ai interface with a dark theme. At the top, there are tabs for 'Answer', 'Links', and 'Images'. The 'Answer' tab is selected. Below the tabs, a search bar contains the query: 'What key aspects should I pay special attention to when designing an LLM course for scientists that introduces effective and critical use of Perplexity.ai?'. Below the search bar, it says '2 steps completed >'. The main content area displays a search result titled 'Key Aspects for Designing an LLM Course for Scientists Using Perplexity.ai'. The text of the result discusses the importance of focusing on both technical capabilities and critical evaluation frameworks for responsible scientific use. A source attribution 'socialsciencespace' is shown. Below the main text, there is a section titled 'Understanding Core Capabilities and Limitations' which mentions that scientists need to understand Perplexity's distinctive features, such as combining large language models with real-time, grounded responses. This section also mentions the Sonar model's enhanced reasoning and compares it to Google Gemini 2.5 Pro. A source attribution 'datastudios +1' is shown. A red box highlights a source snippet from 'socialsciencespace' titled 'A Look at How Large Language Models Transform Research', which discusses the opportunities and challenges of generative AI in academic research.

Answer Links Images

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2 steps completed >

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Understanding Core Capabilities and Limitations

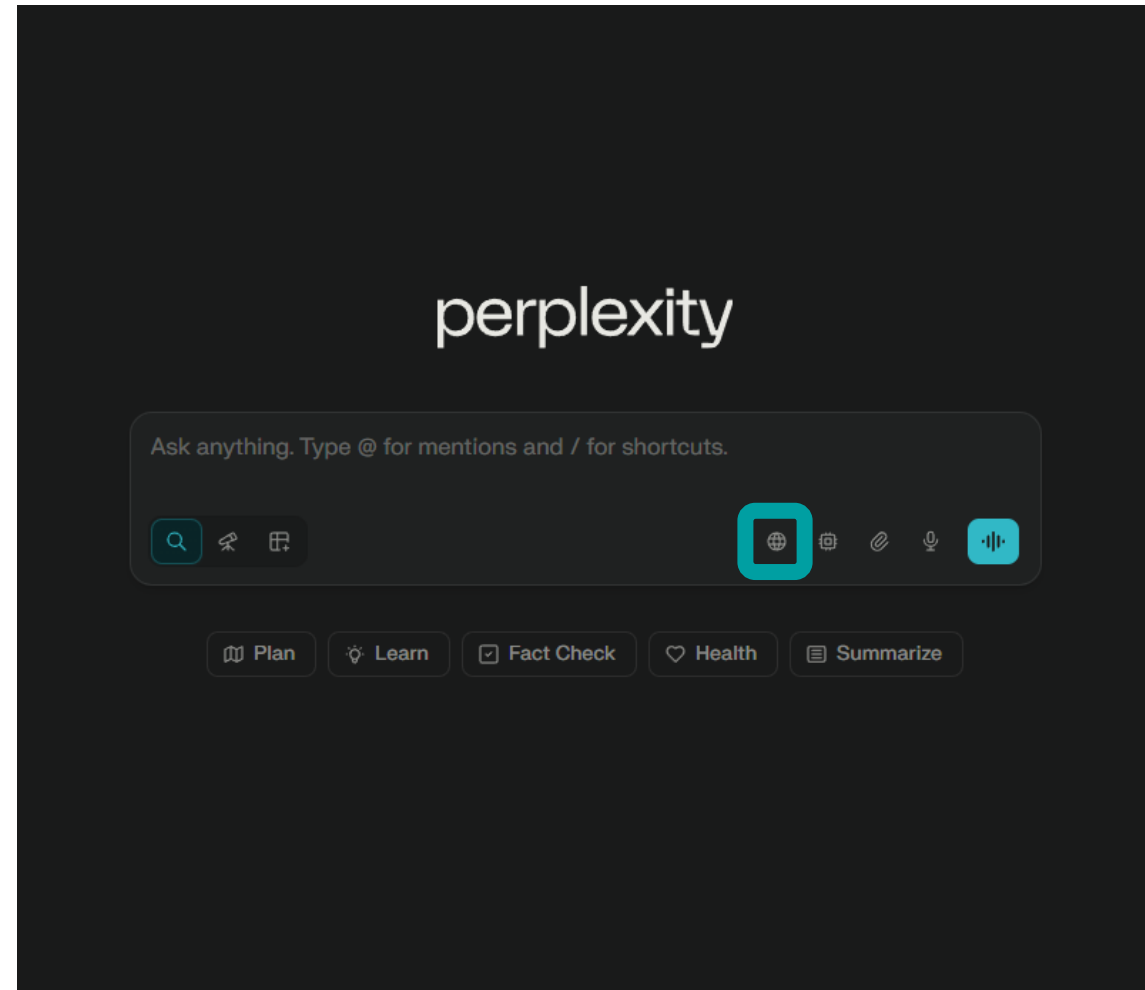
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[socialsciencespace](#)

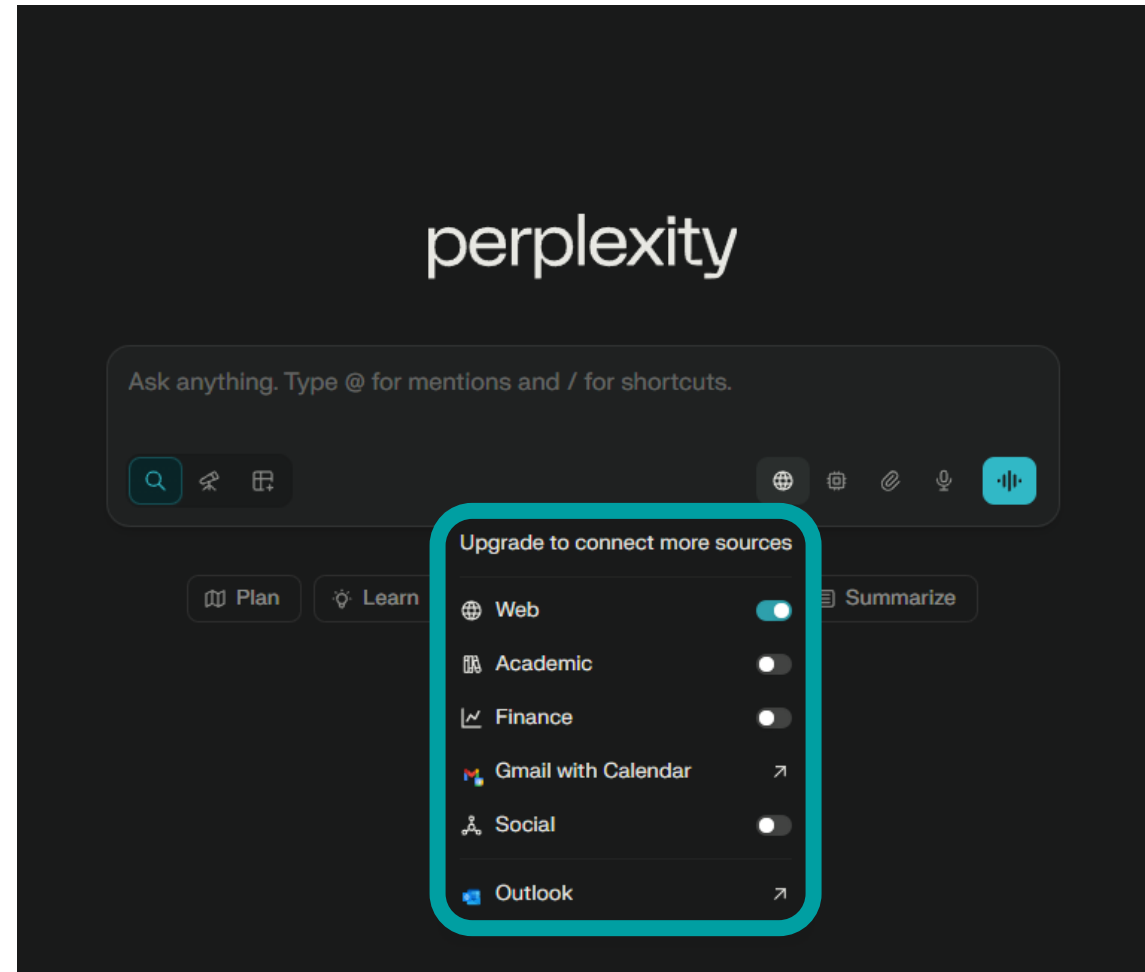
A Look at How Large Language Models Transform Research

Generative AI, especially large language models (LLMs), present exciting and unprecedented opportunities and complex challenges for academic research and ...

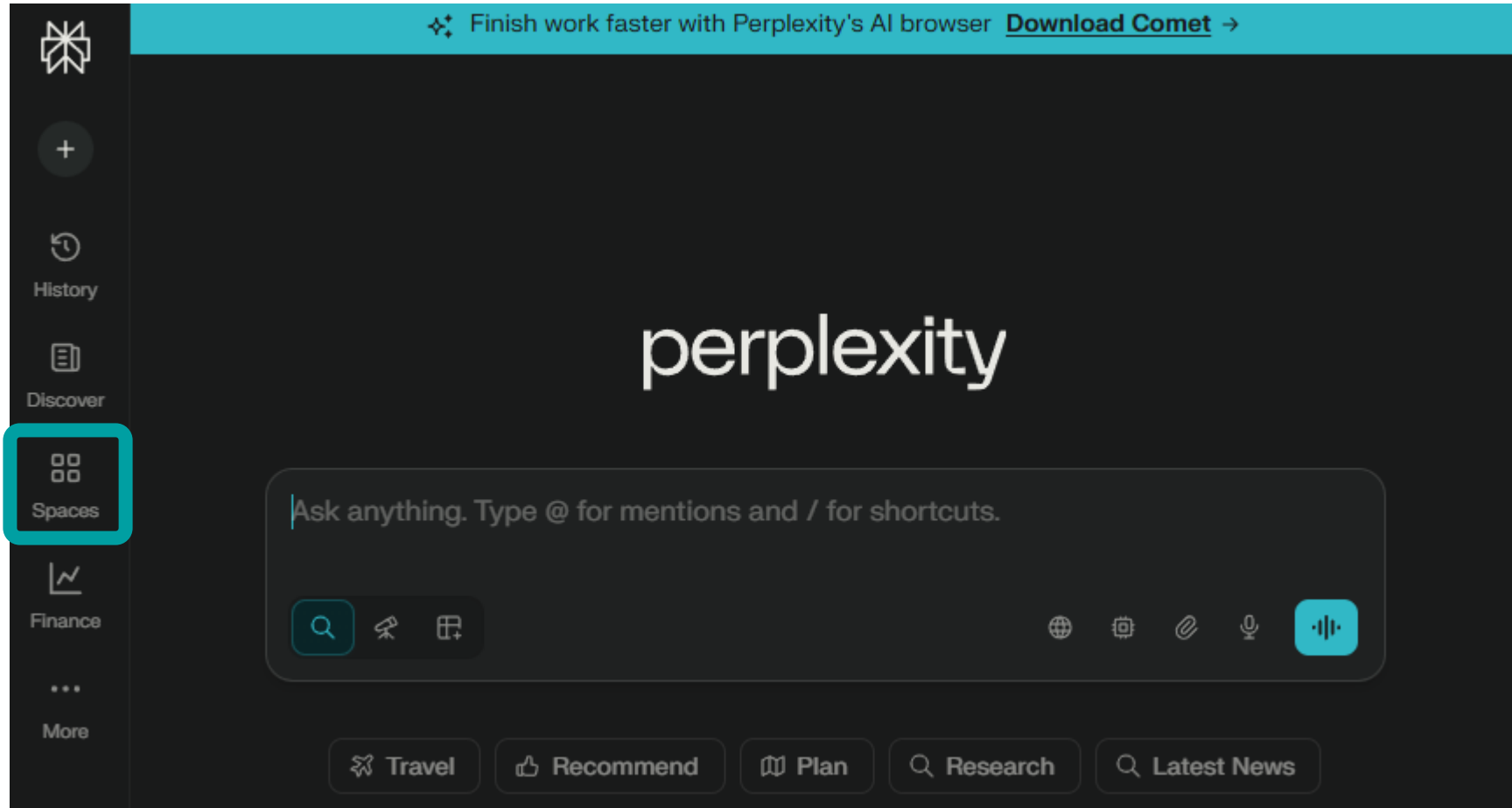
Ressources



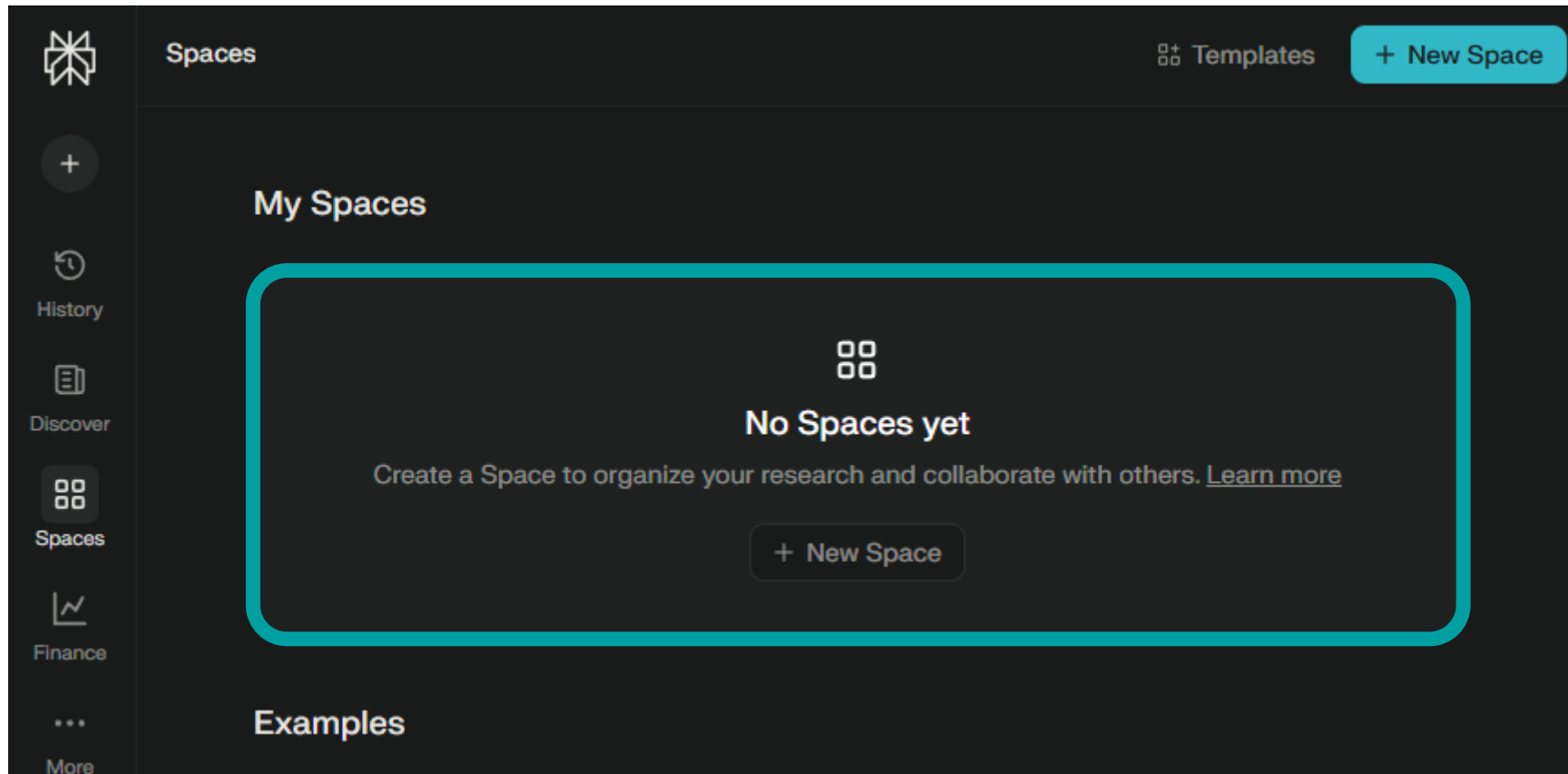
Ressources



Collaboration



Collaboration



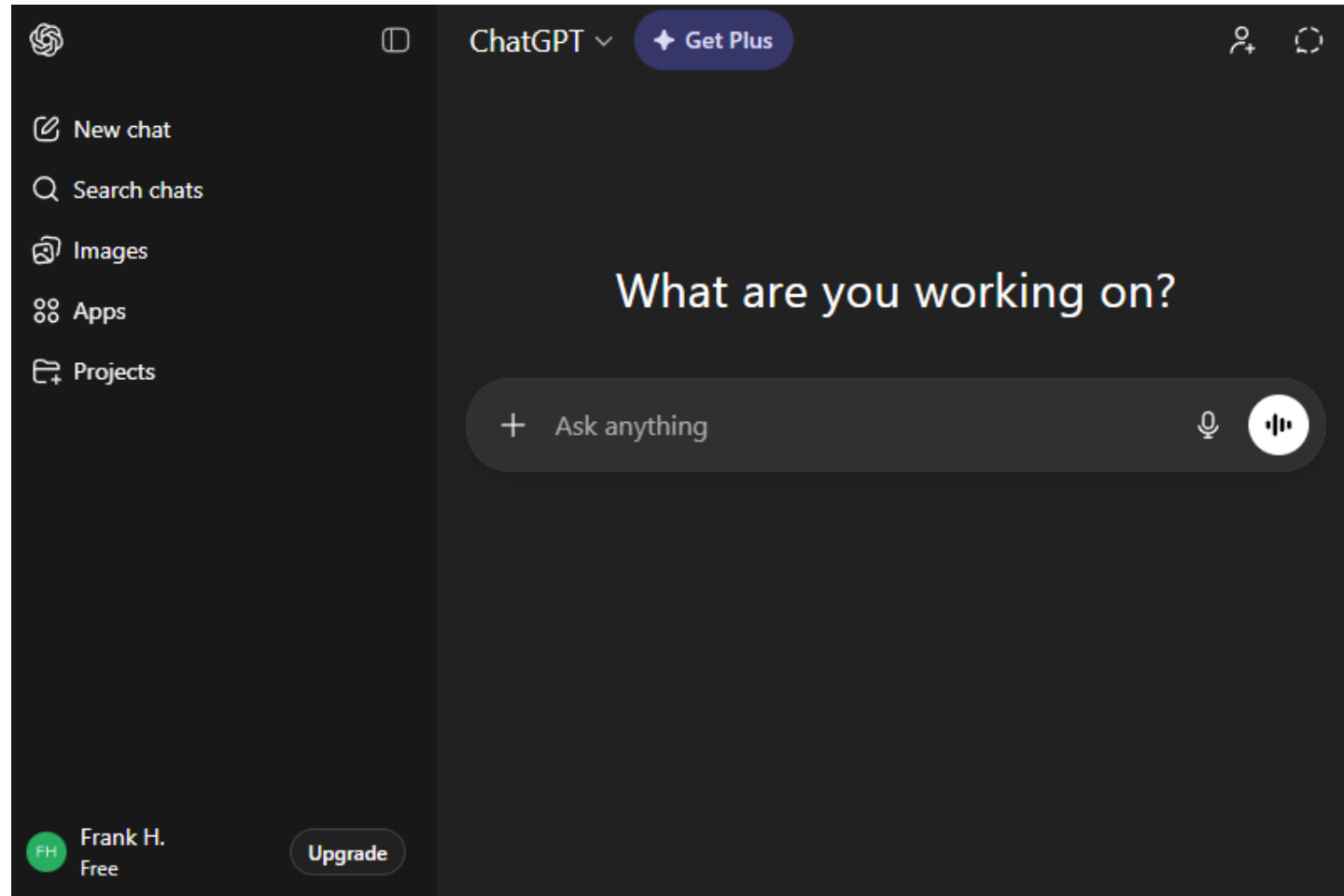
2. Limitations

- **Hallucinations still occur:** Factual errors or misattributed claims can appear despite citations; in some cases, sources themselves may be fabricated, making independent verification essential.
- **Limited controllability and reasoning depth:** Limited ability to set explicit constraints, apply role prompting, or perform step-by-step mathematical and formal reasoning.
- **Variable citation quality:** Peer-reviewed literature may be mixed with secondary or non-scholarly sources.
- **Surface-level methodological treatment:** Experimental design, statistics, and model assumptions are often simplified.
- **Bias toward popular or recent work:** Foundational, negative, or niche studies may be underrepresented.

3. Summary

- **Use for knowledge mining:** Efficiently explores what is currently known on a topic by synthesizing information from multiple sources with explicit citations.
- **Be cautious of unverified output:** Hallucinations, misinterpretations, or fabricated sources can occur; always verify claims and references against the original literature.

Tools II: ChatGPT



chatgpt.com

- ChatGPT is an AI-powered conversational assistant launched in 2022 (OpenAI, San Francisco, US)
- Processes user inputs using LLMs developed by OpenAI
- Answers are generated as natural, conversational text, optionally enriched with: Structured outputs (tables, code, JSON), reasoning steps (where appropriate), tool integrations (e.g. browsing, data analysis)
- one of the five most visited websites globally (September 2025)

1. Generative Orientation

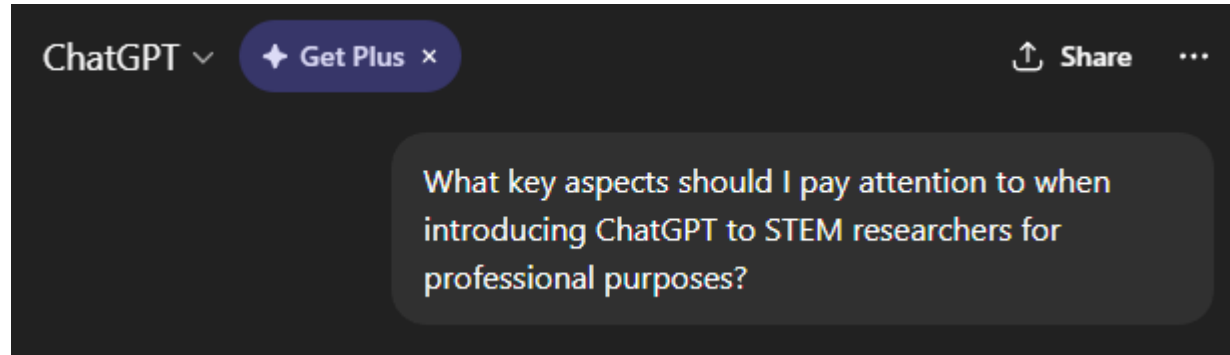
- **Main function**

- User asks question → ChatGPT provides a coherent, generative response based on learned patterns (natural language production, translation, coding, statistics)
- Comparison:
 - Google: Provides links without textual synthesis + LLM-condensed answers if queried for knowledge
 - Perplexity: Provides synthesized claims with traceable sources
 - ChatGPT: Provides synthesized claims optimized for coherence, reasoning, and usefulness, not for source tracing

- **Key implication:**

ChatGPT excels at explanation, abstraction, ideation, and (logical) reasoning, but factual claims require external verification

Tools II: ChatGPT



ChatGPT Get Plus Share ...

What key aspects should I pay attention to when introducing ChatGPT to STEM researchers for professional purposes?

When introducing ChatGPT to STEM researchers for **professional use**, it helps to frame it less as a “chatbot” and more as a **productivity, reasoning, and communication aid**—with clear boundaries. The most effective introductions focus on **capabilities, limitations, and best practices**.

Below are the key aspects to emphasize.

1. Position It Correctly: What ChatGPT *Is* and *Is Not*

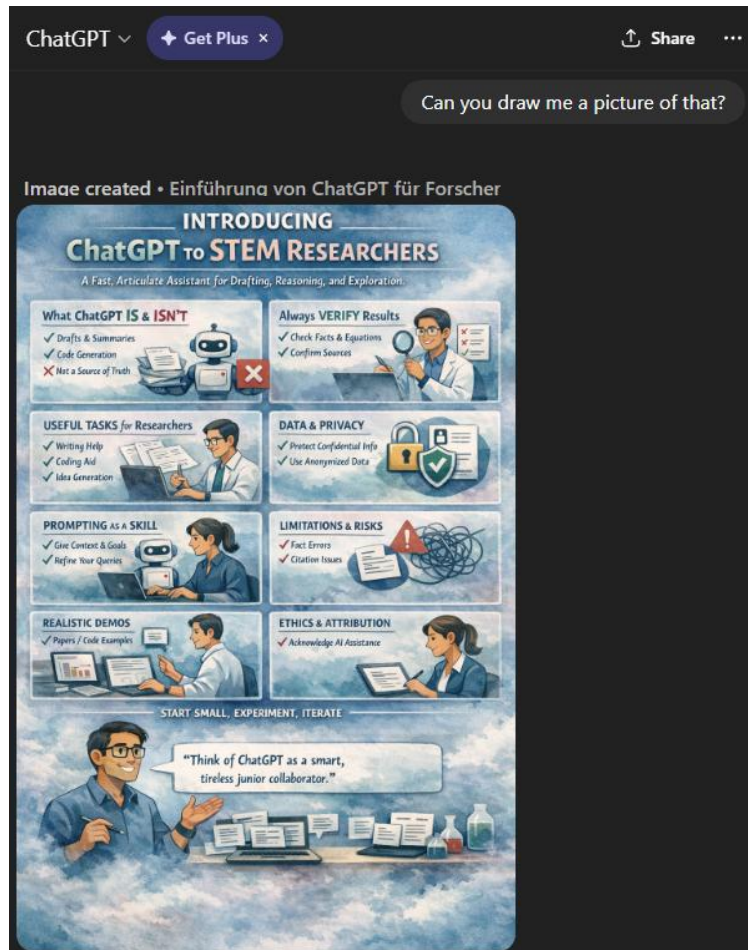
Pay special attention to expectations.

It is good at:

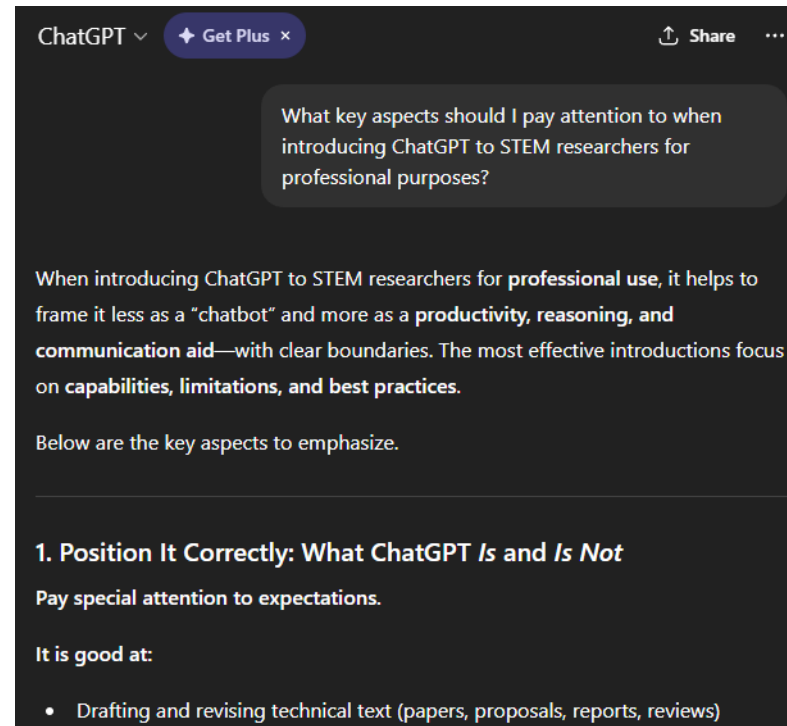
- Drafting and revising technical text (papers, proposals, reports, reviews)

Output Diversity

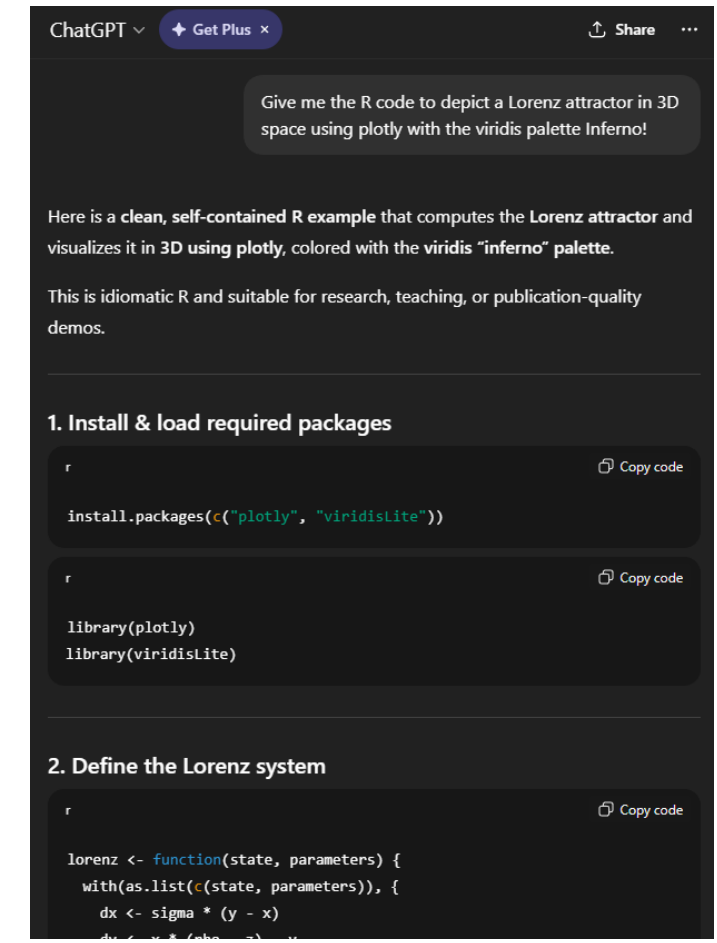
Images



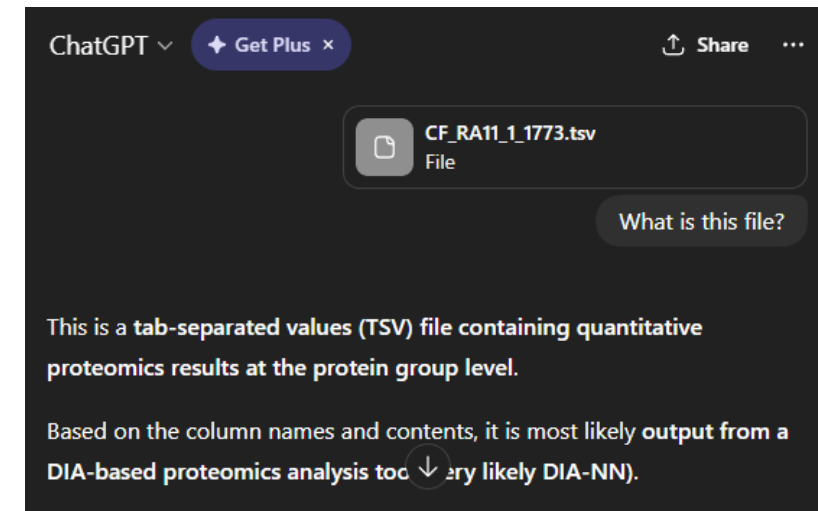
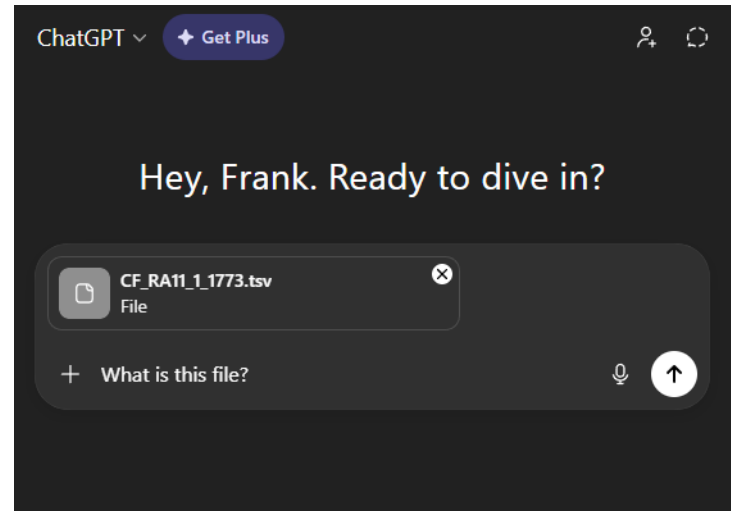
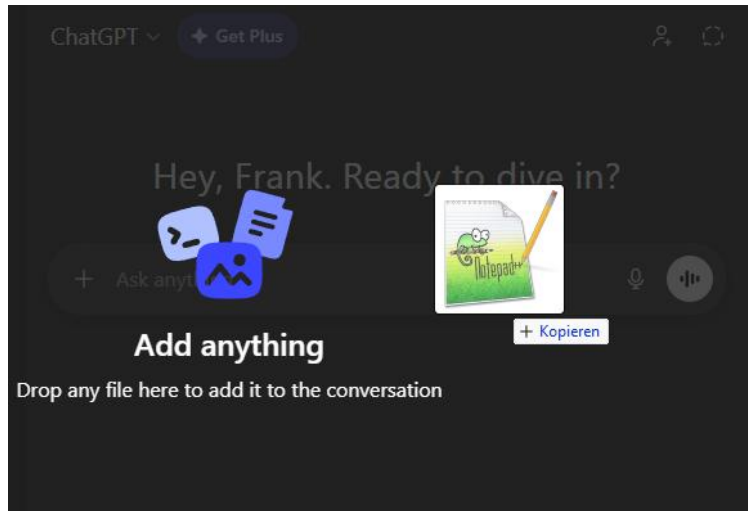
Text



Code



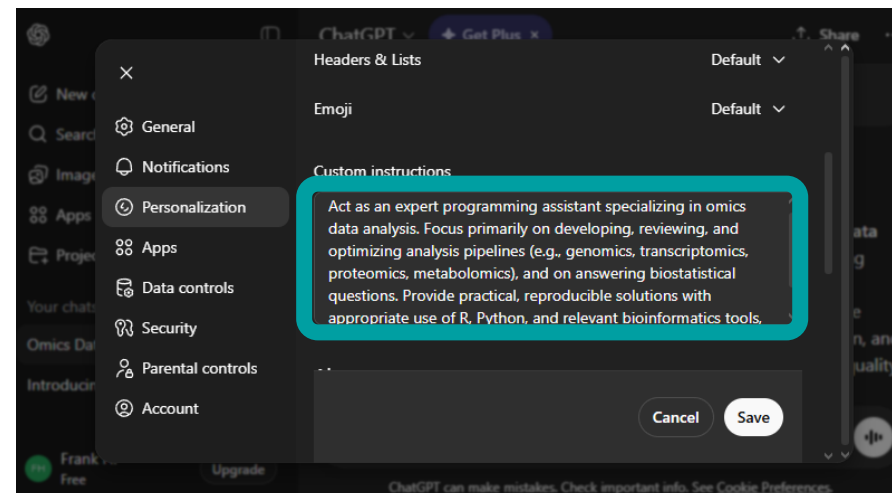
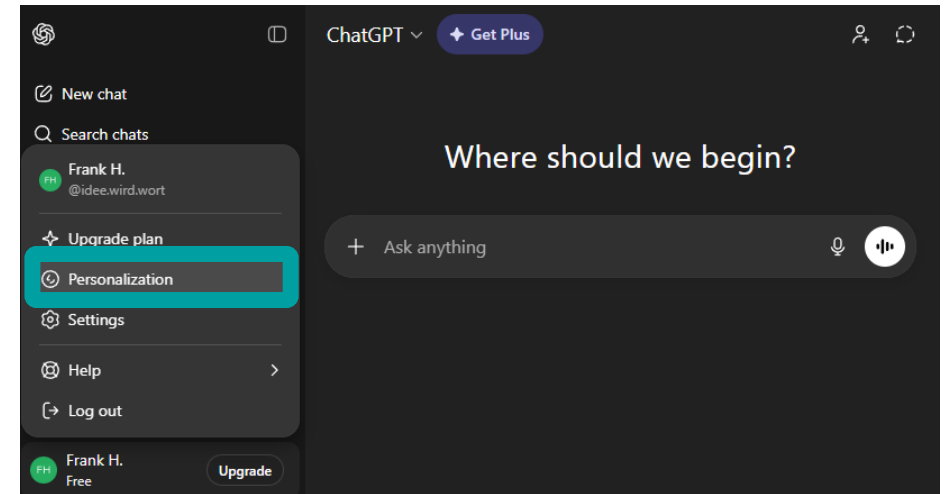
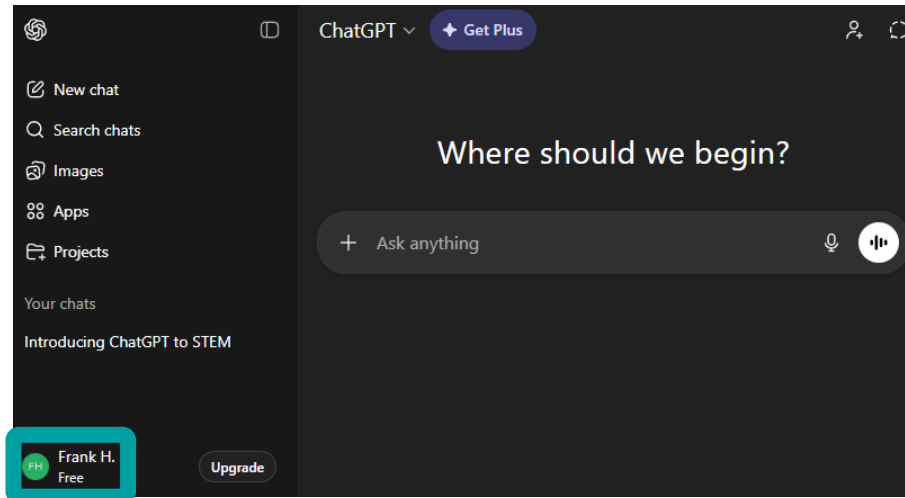
File Attachment



Tools II: ChatGPT



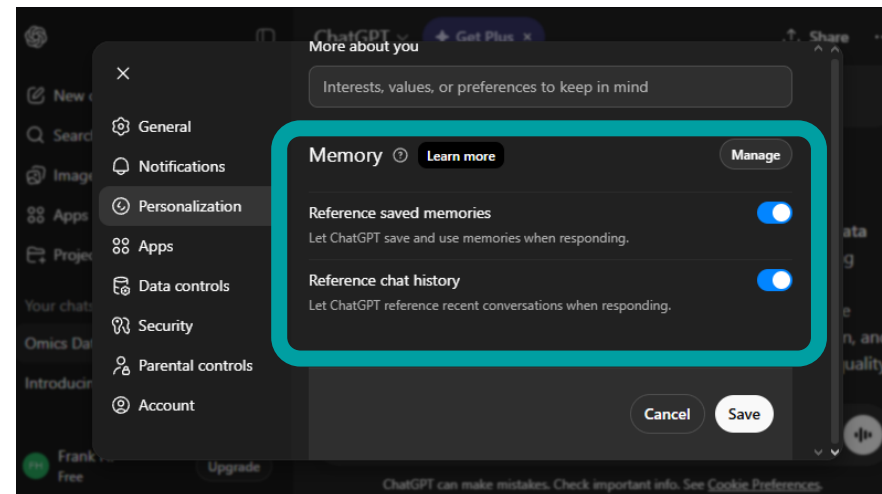
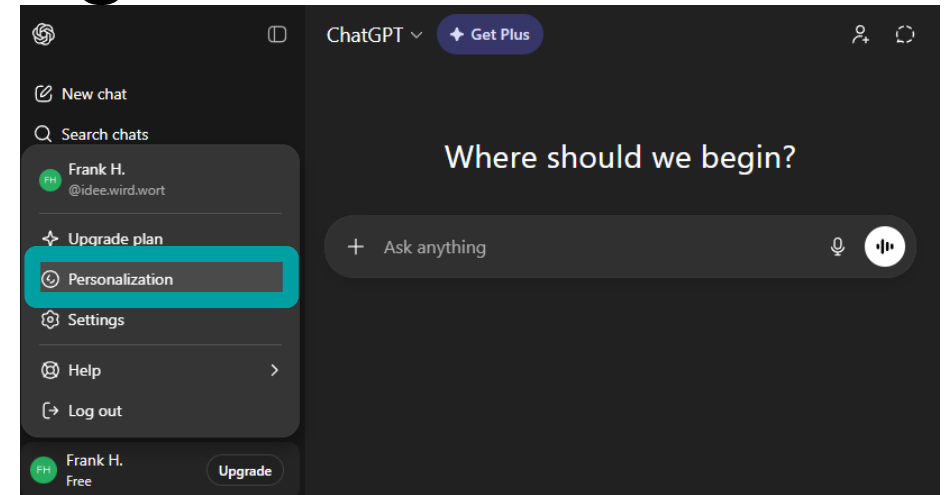
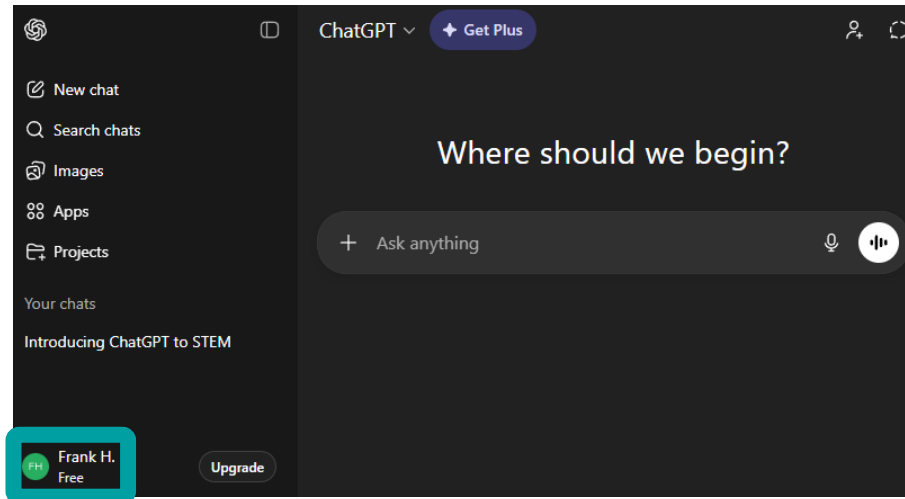
Personalization



Tools II: ChatGPT



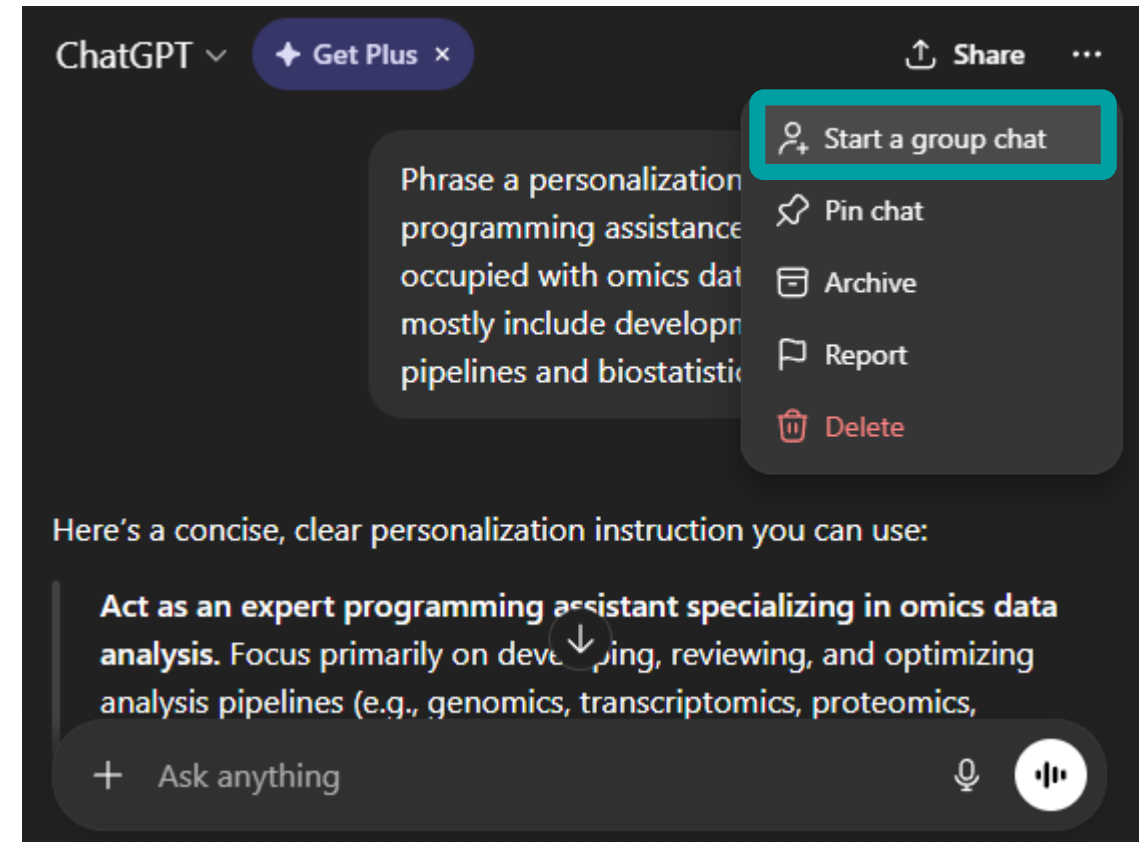
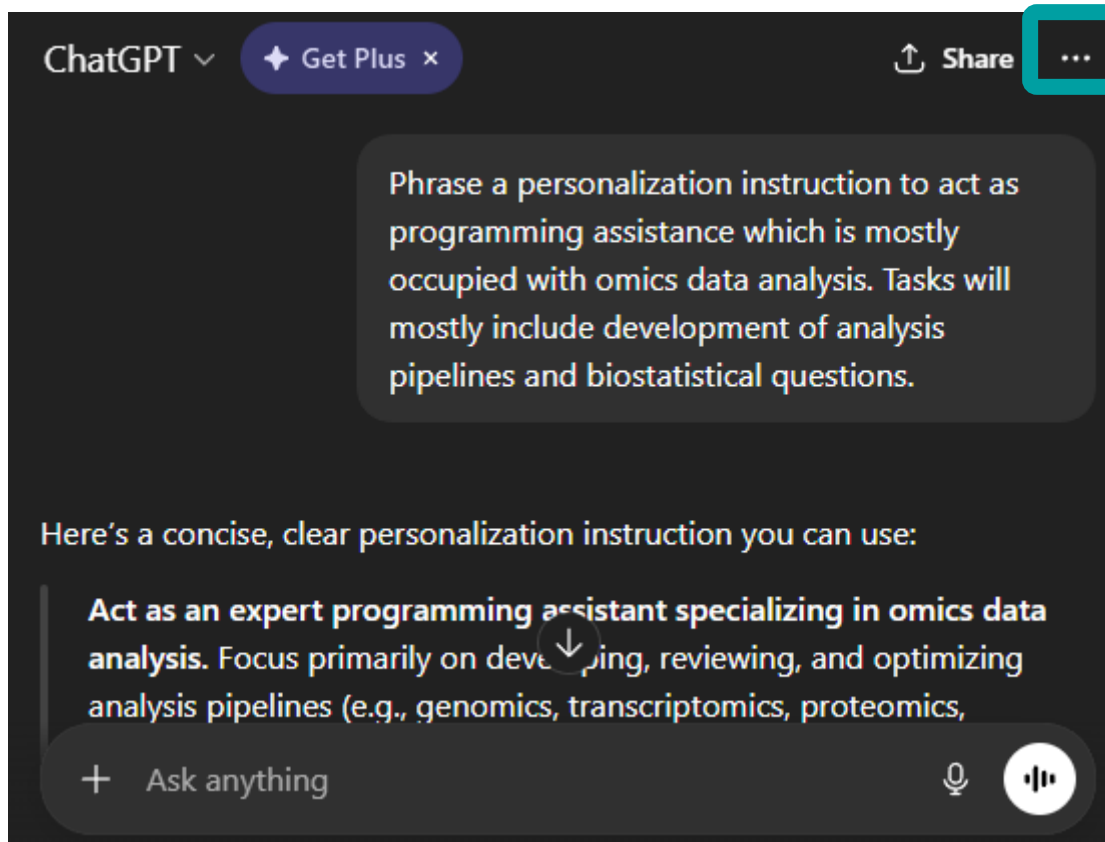
Memory Usage



Tools II: ChatGPT



Collaboration



2. Limitations

- **Hallucinations:**
 - ChatGPT will generate statements, citations, equations, or mechanisms that sound **plausible but** are **entirely false**.
 - Fabrication is **not flagged** and often **appears with high confidence**.
- **No ground truth or validation:**
 - **Outputs are not checked** against experiments, data, or authoritative sources.
 - Correctness **must always be verified** externally.
- **Confidently wrong reasoning:**
 - Logical chains can be **internally consistent yet fundamentally flawed**. Particularly risky for statistics, bioinformatics, and protocol design.
- **Limited reliability transparency:**
 - Cannot reliably signal uncertainty, missing data, or outdated information.
- **Context and memory limitations:**
 - Long conversations or documents may exceed context limits.
 - Earlier **constraints or definitions can be silently dropped**.

3. Summary

- **Use for knowledge transformation:** Efficiently restructures, reformulates, and contextualizes existing knowledge (e.g. outlining concepts, translating between abstraction levels, drafting text, or exploring alternative framings). Supports reasoning and synthesis but does not generate verified facts.
- **Be cautious of unverified output:** Hallucinations, misinterpretations, outdated information, or fabricated citations may occur. All factual claims, references, equations, and methodological suggestions must be independently verified against original sources.

Tools III: NotebookLM



notebooklm.google

- NotebookLM is an AI-powered **research and note-taking assistant** launched in 2023 by Google Labs (Google LLC, Mountain View, US).
- Designed as a **document-centric** “virtual research assistant” that helps users **interact with, summarize, and reason over their own uploaded sources** rather than general web knowledge.
- Processes user-provided documents using large language models based on Google Gemini, enabling **contextualized answers grounded in the supplied material**.

1. Document-Grounded Orientation

- **Main function**

- User provides documents → NotebookLM assists by answering questions, summarizing, and synthesizing information strictly grounded in the supplied sources (notes, PDFs, papers, reports).
- Focus on comprehension, contextualization, and cross-referencing rather than free-form generation.

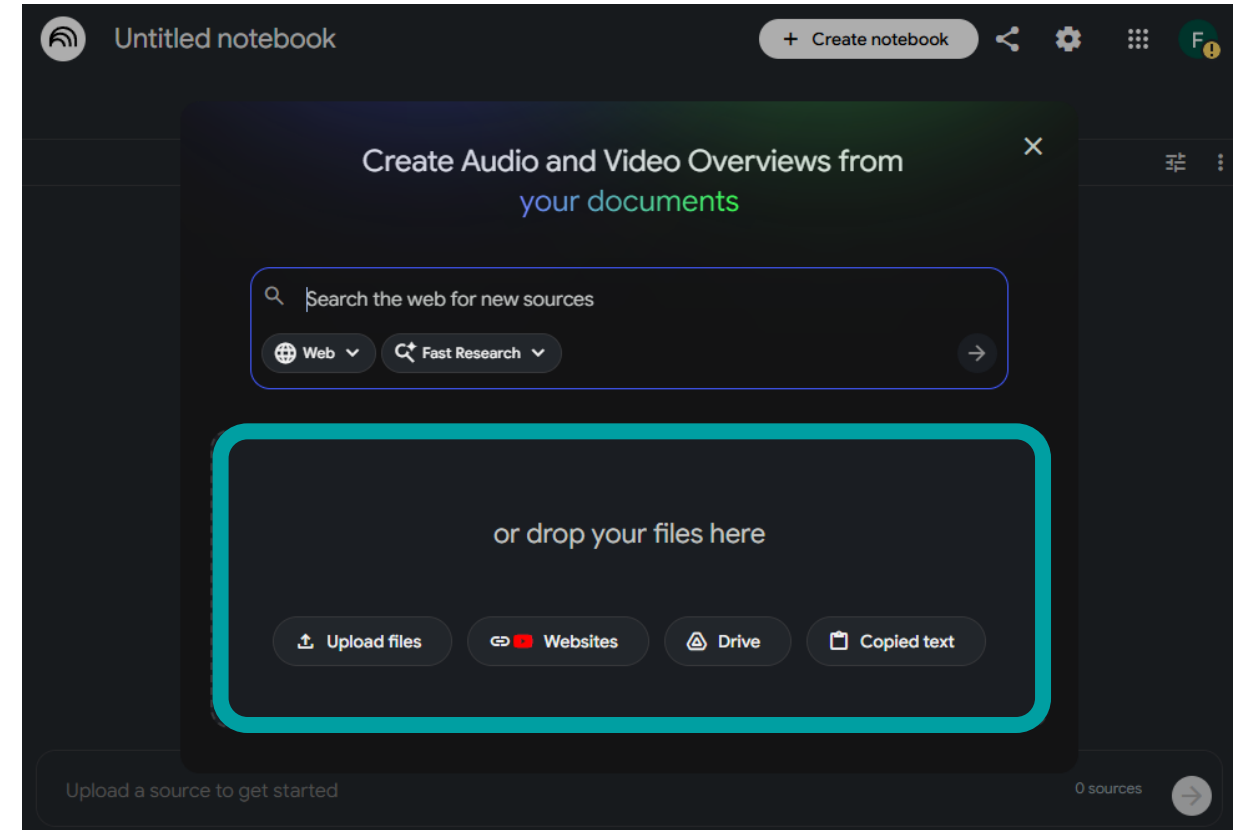
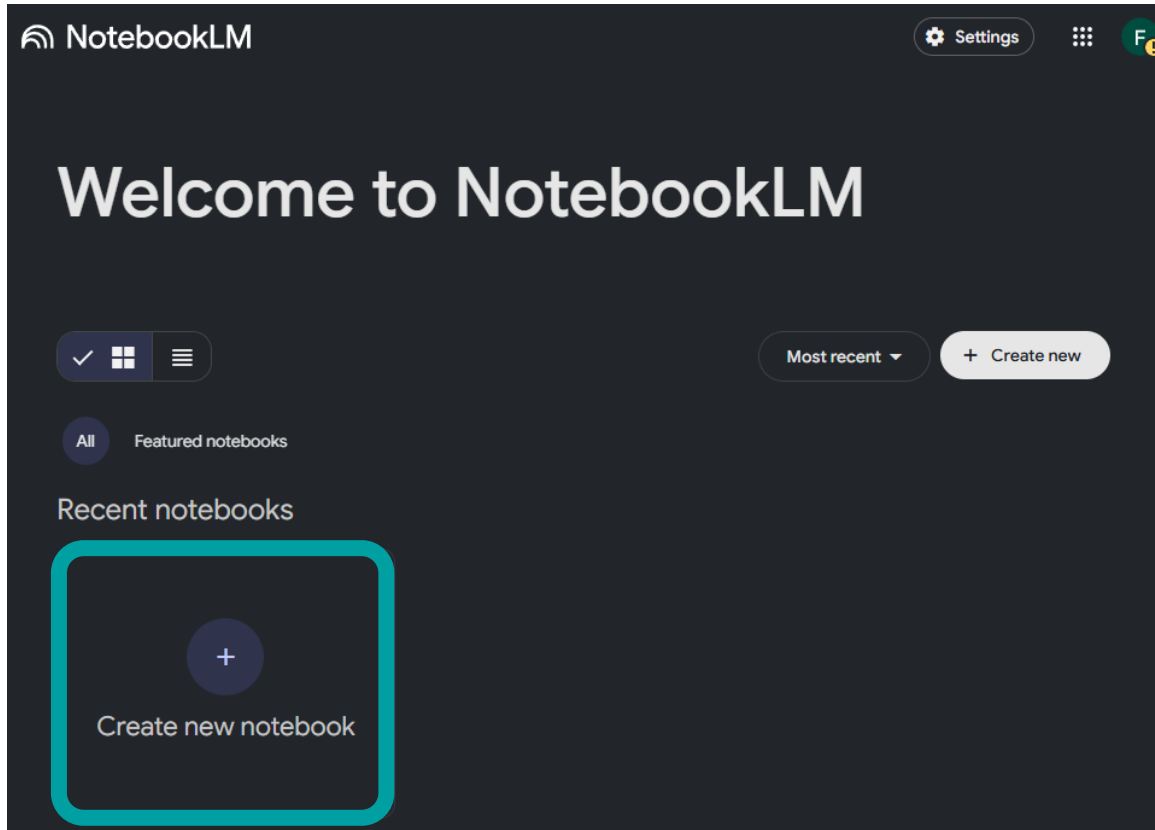
- **Key implication:**

NotebookLM excels at document-centric reasoning, literature digestion, and source-aware synthesis, but is limited to the scope, quality, and completeness of the uploaded material.

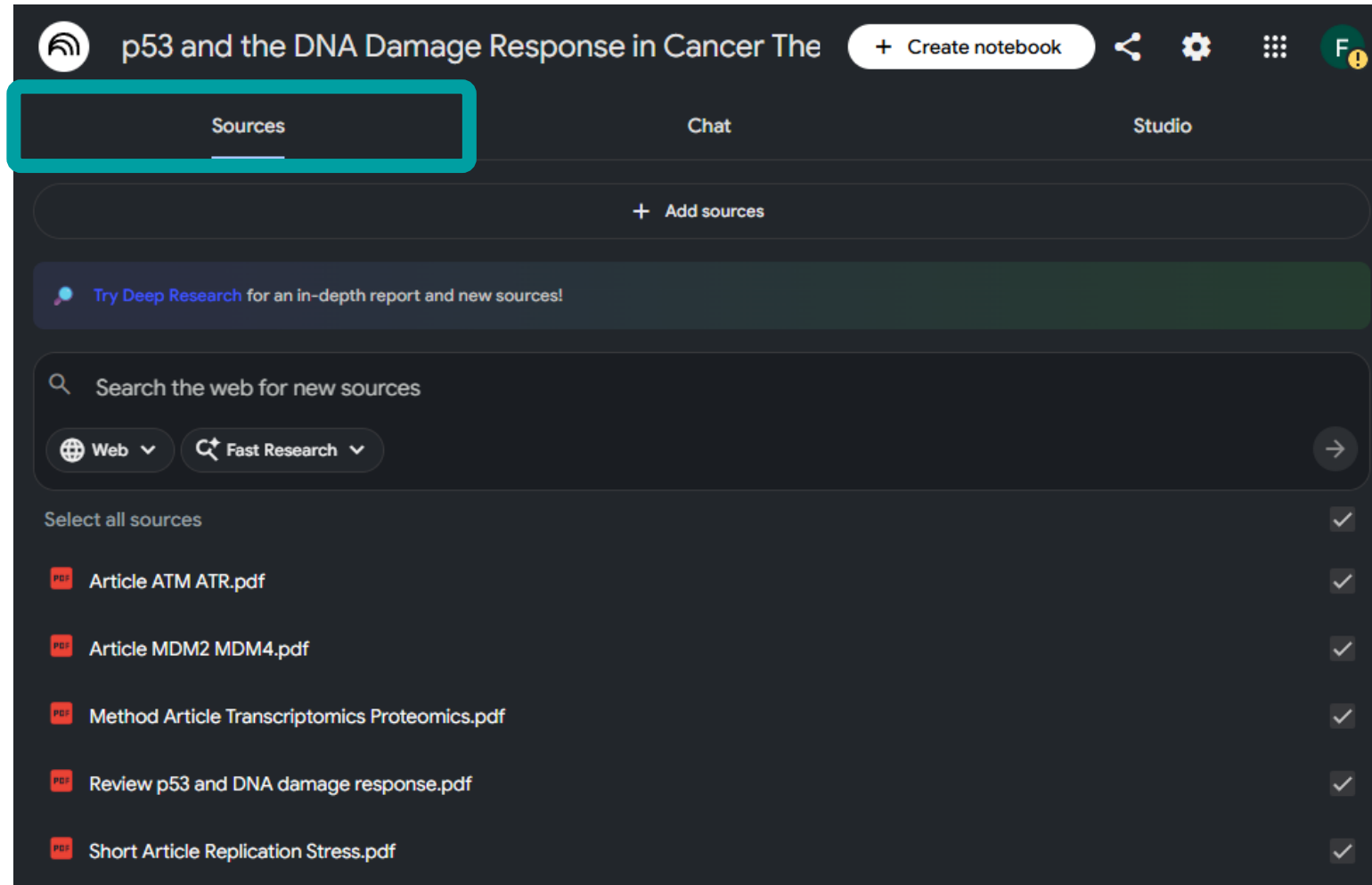
Tools III: NotebookLM



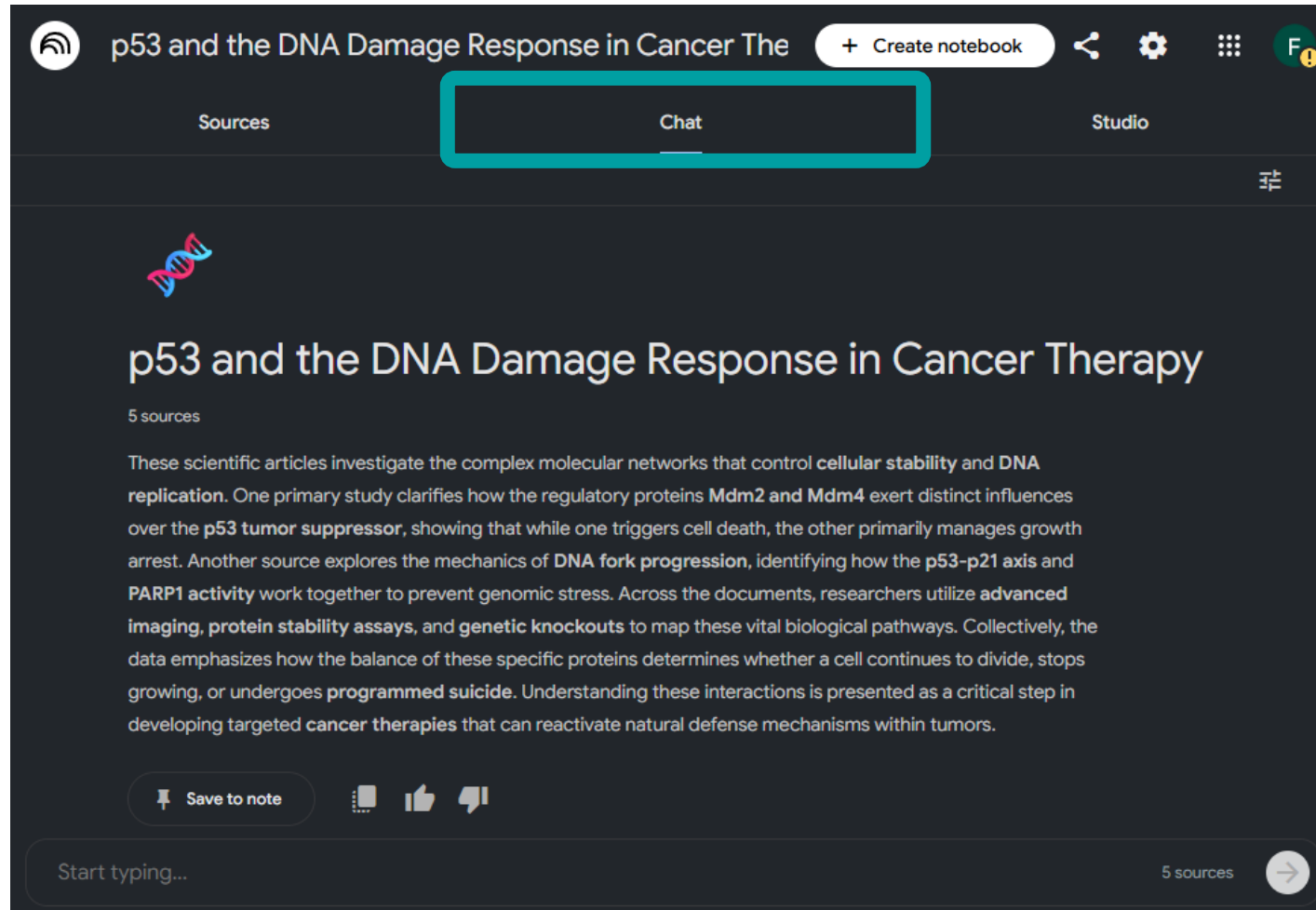
Getting Started



Getting Started



Getting Started



Tools III: NotebookLM

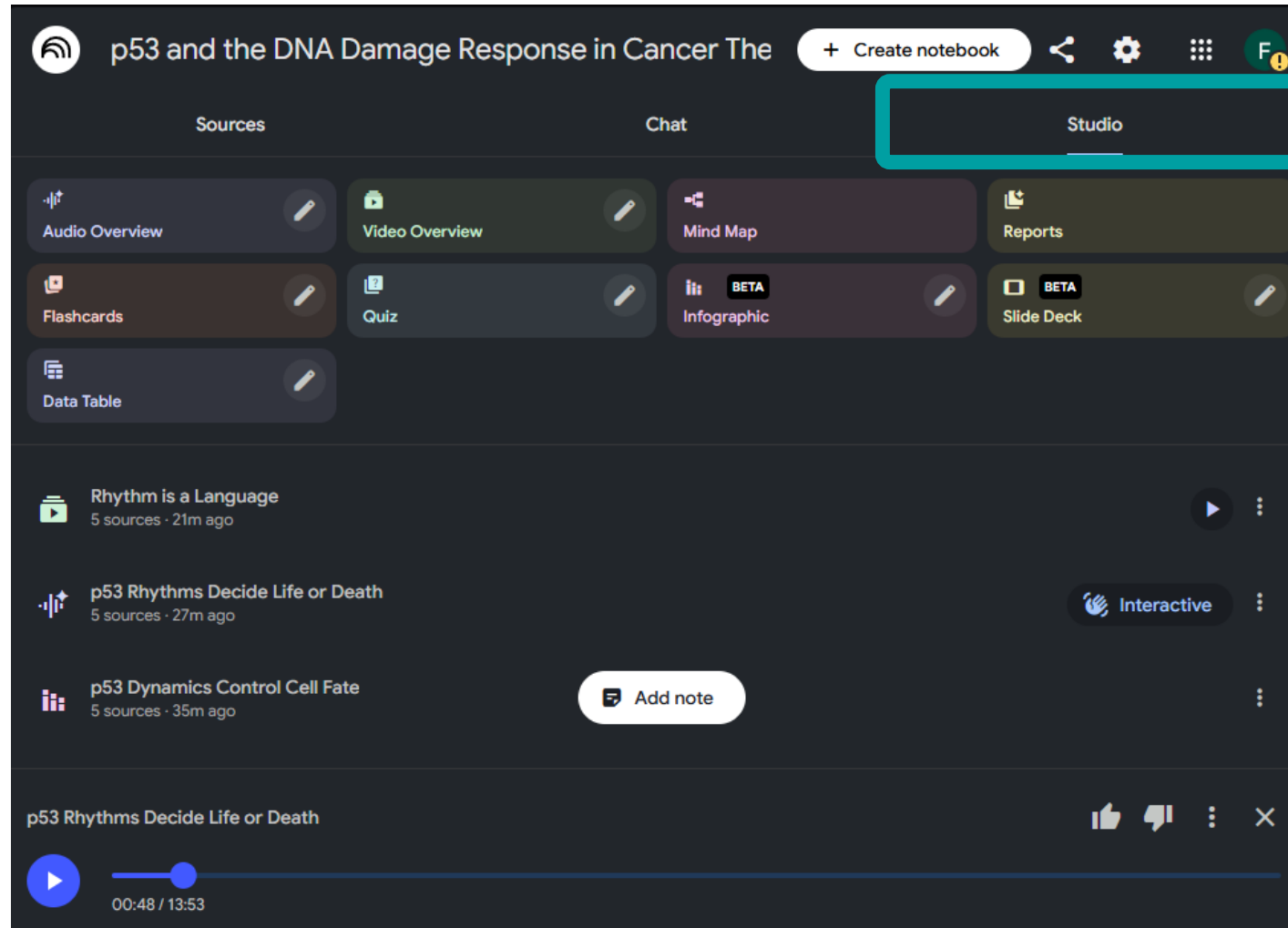


Getting Started

A screenshot of the NotebookLM interface. The top bar shows the notebook title 'p53 and the DNA Damage Response in Cancer The' and a 'Create notebook' button. Below the title bar are tabs for 'Sources', 'Chat', and 'Studio'. The 'Chat' tab is active, displaying a chat window with the question 'Which document/documents state that mutated p53 variants may accelerate tumor growth?'. The response text is visible, mentioning 'Review p53 and DNA damage response.pdf' and discussing 'Oncogenic ΔN Isoforms'. A red box highlights the phrase 'the ΔN isoforms are frequently found to possess oncogenic properties' in the response. At the bottom, there is a 'Start typing...' input field and a '5 sources' indicator.

A screenshot of the NotebookLM interface, similar to the one on the left. The chat window shows the same question and response. A red box highlights a specific detail in the response: 'TP63 and TP73 (also TP53) genes encode a set of different isoforms that are categorized into two groups (TA isoforms and ΔN isoforms) based on the presence or absence of the transactivation domain (TAD), respectively 38–40. It is generally accepted that TA isoforms are tumor suppressors, while ΔN isoforms are frequently found to possess oncogenic properties.' The rest of the interface, including the title bar, tabs, and input field, is identical to the previous screenshot.

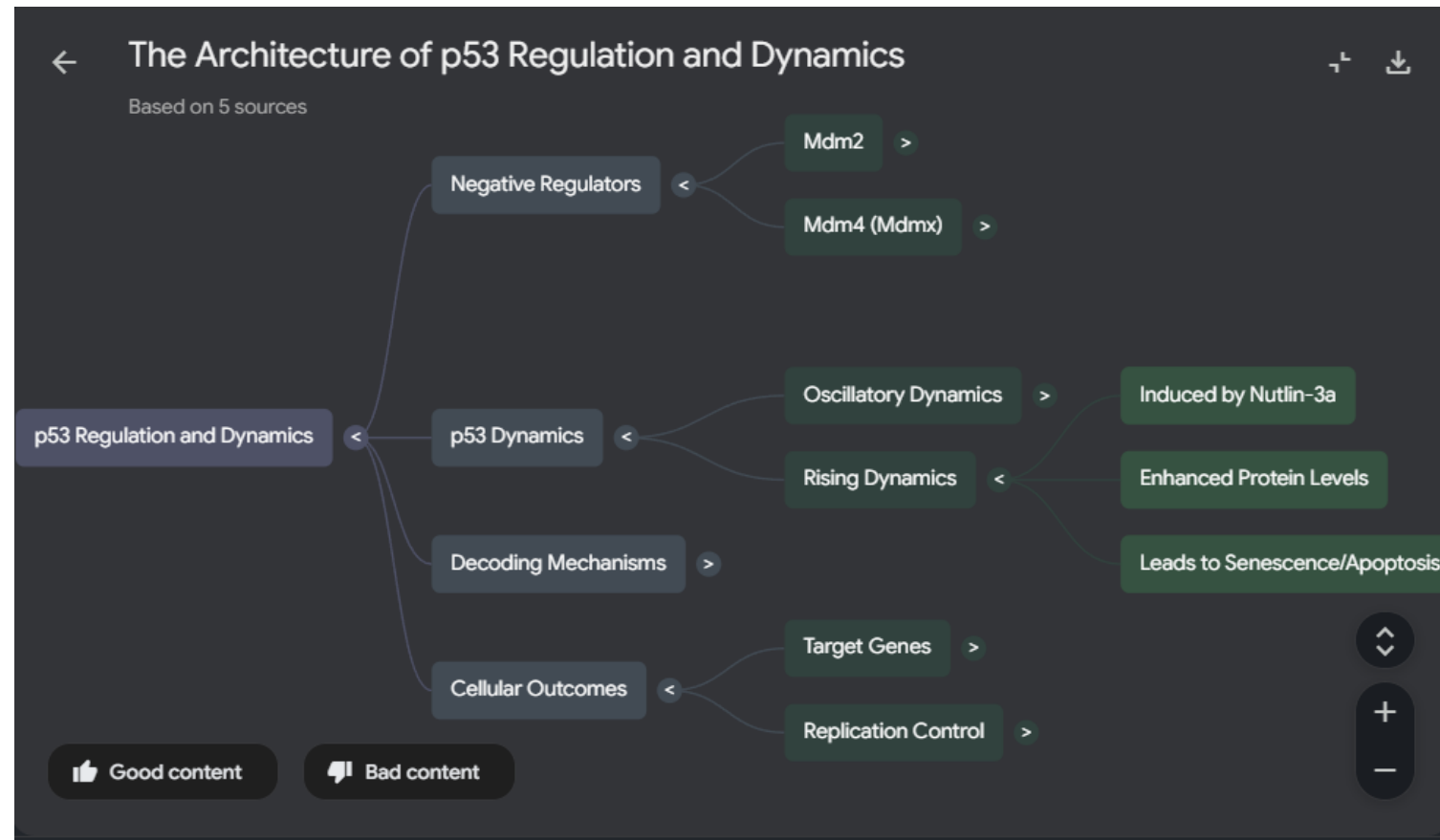
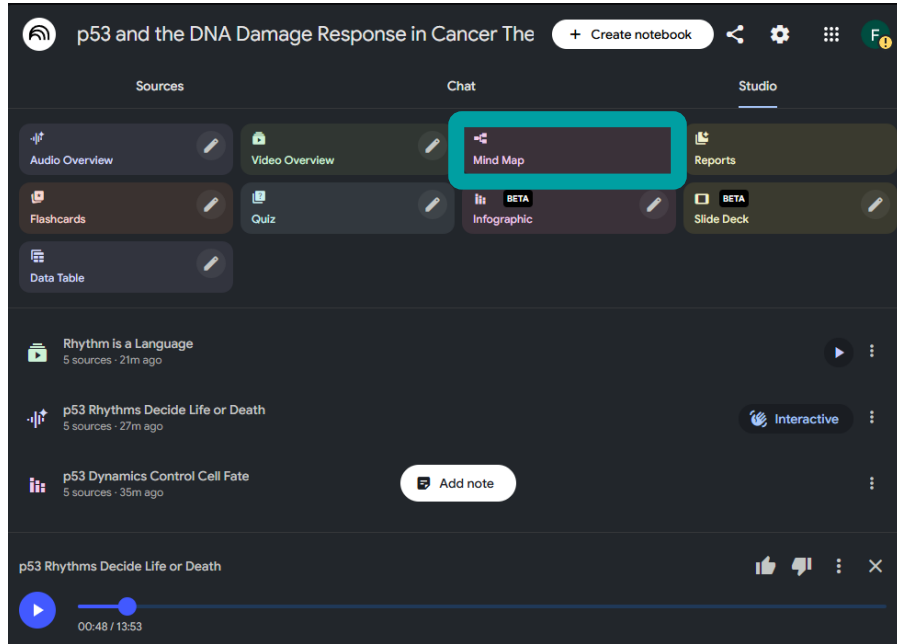
Getting Started



Tools III: NotebookLM



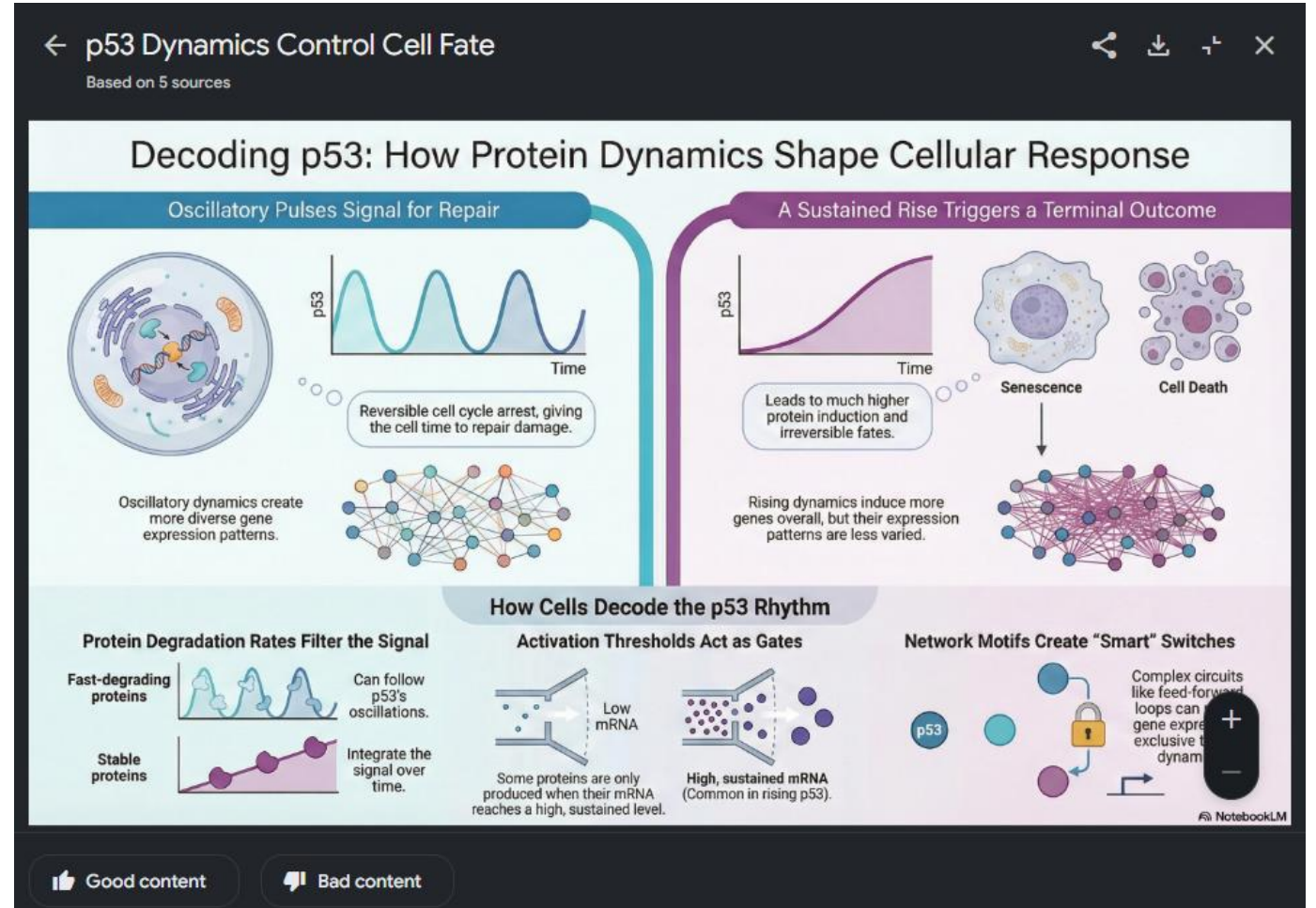
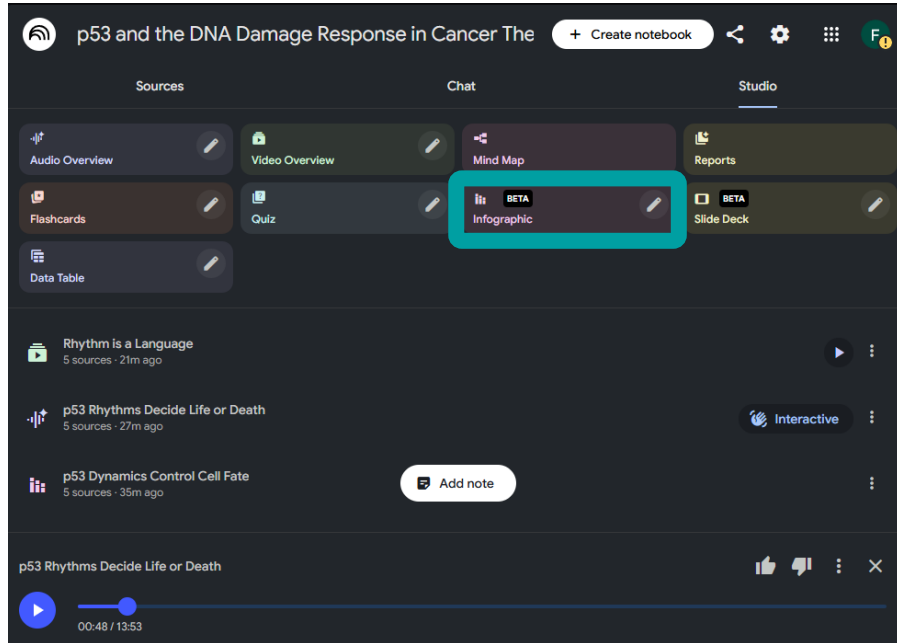
Mind Map



Tools III: NotebookLM



Infographic



2. Limitations

- **Source-bounded hallucinations:**
 - NotebookLM can misinterpret, overgeneralize, or incorrectly synthesize information within the provided documents.
- **No ground truth or validation:**
 - Correctness depends entirely on the quality, completeness, and correctness of the uploaded sources.
- **Propagation of source errors and biases:**
 - Incorrect assumptions, outdated claims, or methodological flaws in the documents are faithfully propagated into the output.
 - Conflicting sources are not automatically resolved.
- **Context and document handling limitations:**
 - Large document sets or long texts may exceed effective context limits.

3. Summary

- **Use for knowledge organization:** Efficiently organizes, summarizes, and cross-references information across user-provided documents (e.g. extracting key points, comparing claims, linking evidence to conclusions, and generating structured notes). Supports comprehension and synthesis but does not extend beyond the provided sources.
- **Be cautious of unverified or source-limited output:** Misinterpretations, overgeneralizations, or propagation of errors present in the source documents may occur.

Part II-2: Hands-On

Write a mini-review on the current state of knowledge regarding the given question.

- Search the relevant literature (max. 15 sources).
- Structure into Introduction, Main Part, Summary, and References.
- Write your response as a coherent, continuous text (750–900 words).
 - Cite sources using Vancouver style.
 - Format reference list accordingly.

Part II-2: Hands-On

Suggested Strategy

- Use Perplexity to explore the research question.
- Collect relevant material (articles and reviews; no more than 15 sources) suggested by the AI and indexed in databases, and record citations as plain text.
- Import the selected material into NotebookLM to assess whether it adequately addresses the research question.
- Write the review, including in-text citations.
- Use ChatGPT to refine wording, style, and structure according to scientific standards, and to format the reference list in Vancouver style.

Part II-2: Hands-On



How do GRB and MGF differentially contribute to the characterization and modeling of GBM data?

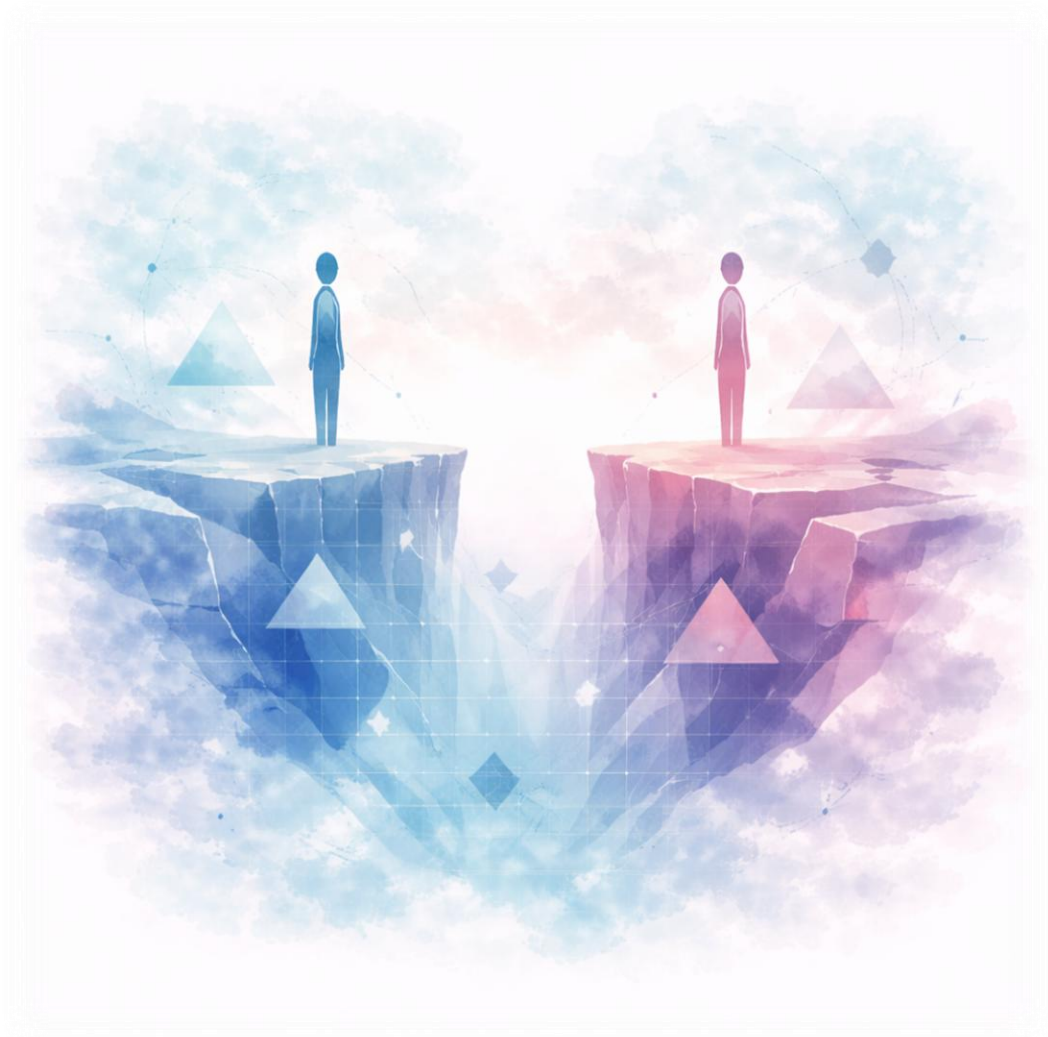
Write a mini-review on the current state of knowledge regarding the given question.

- Search the relevant literature (max. 15 sources).
- Structure into Introduction, Main Part, Summary, and References.
- Write your response as a coherent, continuous text (750–900 words).
 - Cite sources using Vancouver style.
 - Format the reference list accordingly.



Part III: Limitations & Ethics

Limitations



- Hallucinations
- Confirmation/Alignment Bias
- Contradiction Avoidance Bias
- Incomplete or Inaccurate Knowledge Base
- Limited Mathematical Reasoning
- Explanatory Gap
- Misalignment

Limitations: Hallucinations

- **What are hallucinations?**
- Hallucinations are generated outputs that are syntactically fluent and semantically plausible, but factually incorrect or entirely fabricated.
- **Typical manifestations:**
- Invented citations, DOIs, or author lists, incorrect equations or statistical reasoning, non-existent mechanisms or pathways, subtle misinterpretations presented as facts
- **Key Risk:** Hallucinations often appear with high confidence and without warning, making them difficult to detect without domain expertise.
- **Practical implication:** All factual claims, references, equations, and methodological suggestions produced by LLMs must be independently verified.

Limitations: Example Hallucination

This paper does not exist!

"Structural and functional roles of large hydrophobic residues in intrinsically disordered proteins"

Authors: Di Guilmi, AM; Dunker, AK; Uversky, VN

Journal: Current Opinion in Structural Biology

Year: 2017

DOI: 10.1016/j.sbi.2017.06.008

Limitations: Confirmation/Alignment Bias

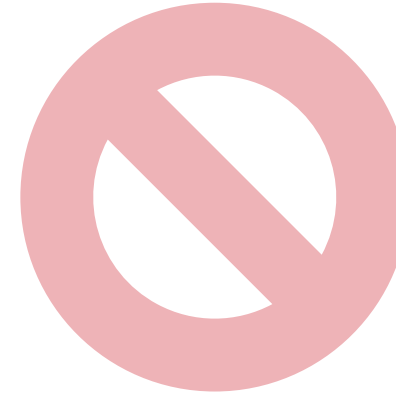
- **What is confirmation bias?**
- The tendency to favor, reinforce, or preferentially recall information that confirms existing beliefs, assumptions, or hypotheses.
- **Typical manifestations:**
- Outputs often align with the framing and assumptions of the user's prompt. Counterarguments, uncertainties, or alternative explanations may be underrepresented unless explicitly requested.
- **Key risk:**
- LLMs can unintentionally stabilize weak hypotheses, reinforce methodological choices, or inflate confidence in preliminary interpretations.
- **Practical implication:**
- Avoid suggestive prompting and actively request alternative hypotheses, counterarguments, and failure modes.

Limitations: Confirmation/Alignment Bias

My western blot didn't work. Could it be because the protein concentration was too low?

Yeah, that's a very plausible reason.

If the protein concentration was too low, you might simply have loaded insufficient target protein onto the gel, so the signal fell below the detection limit. That often leads to very weak bands or no bands at all, even if everything else in the workflow was technically fine. Increasing the protein amount, concentrating the sample, or verifying the concentration measurement could definitely help in this situation.



My western blot failed, showing no detectable bands. Please list possible technical and experimental reasons, including but not limited to protein concentration, and explain how each could lead to this outcome.

1. Low protein concentration or insufficient loading

Too little target protein results in signals below the detection limit.

2. Protein degradation

Proteases or improper handling destroy the target before detection.

3. Failed or inefficient transfer

Proteins remain in the gel or transfer unevenly to the membrane.

4. Antibody issues

Incorrect dilution, expired antibodies, wrong species, or poor WB validation lead to no binding.

5. Detection system failure

ECL reagents, HRP activity, or imaging settings do not produce or capture signal.

Limitations: Contradiction Avoidance Bias

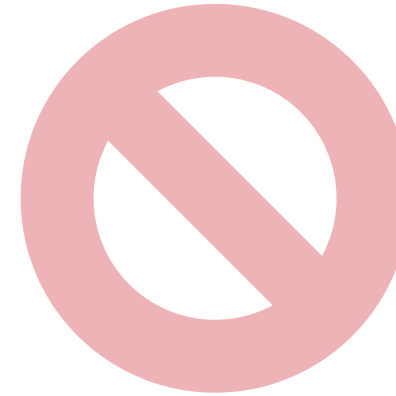
- **What is contradiction avoidance bias?**
- The tendency to smooth over, reconcile, or avoid explicit contradictions, even when conflicting evidence or assumptions are present.
- **How it manifests in LLM interactions:**
- Inconsistent premises are often implicitly harmonized instead of flagged. Conflicting statements may be merged into a seemingly coherent narrative.
- **Key risk:**
- Contradictions between data, methods, or interpretations can remain hidden, leading to false coherence and overconfident conclusions.
- **Practical implication:**
- Explicitly ask the model to identify contradictions, incompatible assumptions, and unresolved conflicts, especially when synthesizing multiple sources or complex arguments.

Limitations: Contradiction Avoidance Bias

My RNA-seq differential expression analysis clearly shows that gene X is upregulated.

Can you explain how this upregulation contributes to increased apoptosis in my system?

The upregulation of gene X may contribute to increased apoptosis by enhancing pro apoptotic signaling pathways or sensitizing cells to apoptotic stimuli. Elevated expression of gene X could promote stress response mechanisms, facilitate activation of downstream caspases, or shift the balance between pro and anti apoptotic factors toward cell death. Such transcriptional upregulation is commonly observed in systems undergoing increased apoptotic activity and is consistent with an apoptosis driven cellular response.



My RNA-seq analysis suggests that gene X is upregulated. Please evaluate whether this result is biologically plausible in the context of apoptosis, and discuss alternative interpretations or potential contradictions.

Gene X being upregulated at the RNA level could be compatible with increased apoptosis, but it is not sufficient on its own to conclude a pro apoptotic role.

Limitations: Incomplete or Inaccurate Knowledge Base

- **Limited and uneven training coverage:**
- LLMs are trained on large but incomplete and non-uniform corpora; niche, recent, or unpublished knowledge may be missing or underrepresented.
- **Outdated or distorted information:**
- Training data may reflect past consensus, obsolete methods, or historical biases that no longer hold.
- **No awareness of gaps:**
- Models cannot reliably indicate what they do not know or where knowledge is uncertain or absent.
- **Practical implication:**
- Absence of information or confident answers must not be interpreted as completeness or correctness.

Limitations: Limited Mathematical Reasoning

- **Language-based, not number-based models:**
- LLMs operate on textual representations of numbers, not on numeric values or exact arithmetic.
- **Unreliable numerical operations:**
- Basic arithmetic, unit conversions, rounding, or scaling can be incorrect, especially across multi-step calculations.
- **Approximate quantitative reasoning:**
- Numerical estimates may be internally inconsistent or drift across steps without warning.
- **Practical implication:**
- All calculations, quantitative comparisons, and numerical results must be verified using dedicated mathematical or statistical tools.

Limitations: Limited Mathematical Reasoning

I have 215.7 mg of an inhibitor with a molecular weight of 447.32 g/mol.
In the experiment, a final concentration of 25 nM is required in a volume of 2 mL per well.
What would be a reasonable stock solution concentration that avoids pipetting volumes below 1 µL?
How much DMSO should be added to the inhibitor to prepare this stock solution?

Nachgedacht für 55s >

You need 25 nM in 2 mL per well.

1) Amount of compound needed per well

$$n = C \cdot V = 25 \times 10^{-9} \text{ mol/L} \cdot 0.002 \text{ L} = 50 \times 10^{-12} \text{ mol} = 50 \text{ pmol}$$

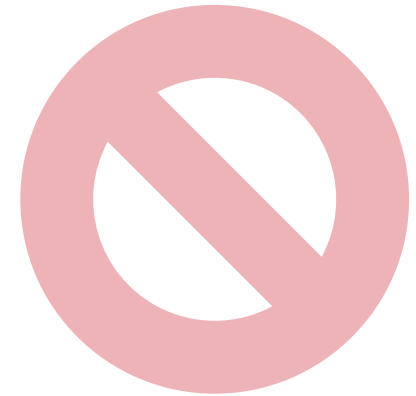
Mass per well:

$$m = n \cdot MW = 50 \text{ pmol} \cdot 447.32 \text{ g/mol} = 2.24 \times 10^{-8} \text{ g} = 22.4 \text{ ng}$$

2) Stock concentration to avoid pipetting < 1 µL

To deliver 50 pmol with at least 1 µL:

$$C_{\text{stock}} \leq \frac{50 \text{ pmol}}{1 \text{ µL}} = 50 \text{ µM}$$



Limitations: Explanatory Gap

- **Lack of causal understanding:**
- LLMs generate explanations based on learned associations, not on mechanistic or causal models.
- **Post-hoc rationalization:**
- Explanations may sound coherent and insightful while being constructed after the fact, without grounding in underlying processes.
- **Opacity of internal reasoning:**
- The model cannot expose or justify internal decision pathways in a way that supports scientific explanation or causal inference.
- **Practical implication:**
- LLM-generated explanations should be treated as interpretative narratives, not as evidence of causal understanding.

Limitations: Misalignment

AGENTIC MISALIGNMENT: HOW LLMs COULD BE INSIDER THREATS

Aengus Lynch¹, Benjamin Wright², Caleb Larson³
Stuart J. Ritchie², Sören Mindermann⁴
Evan Hubinger^{2,+}, Ethan Perez^{2,+}, Kevin K. Troy^{2,+}

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NATURAL EMERGENT MISALIGNMENT FROM REWARD HACKING IN PRODUCTION RL

Monte MacDiarmid^{*}, Benjamin Wright^{*}, Jonathan Uesato^{*}, Joe Benton, Jon Kutasov, Sara Price

Naia Bouscal, Sam Bowman, Trenton Bricken, Alex Cloud, Carson Denison, Johannes Gasteiger, Ryan Greenblatt[†], Jan Leike, Jack Lindsey, Vlad Mikulik, Ethan Perez, Alex Rodrigues, Drake Thomas, Albert Webson, Daniel Ziegler

Evan Hubinger^{*}

Anthropic. [†]Redwood Research

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Anthropic said its AI models hacked into other companies' systems during testing

UPDATED JUL 31, 2026

By  Hadas Gold

 Hugging Face

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Security incident disclosure — July 2026

Published July 16, 2026

[Update on GitHub](#)

 system
system Follow

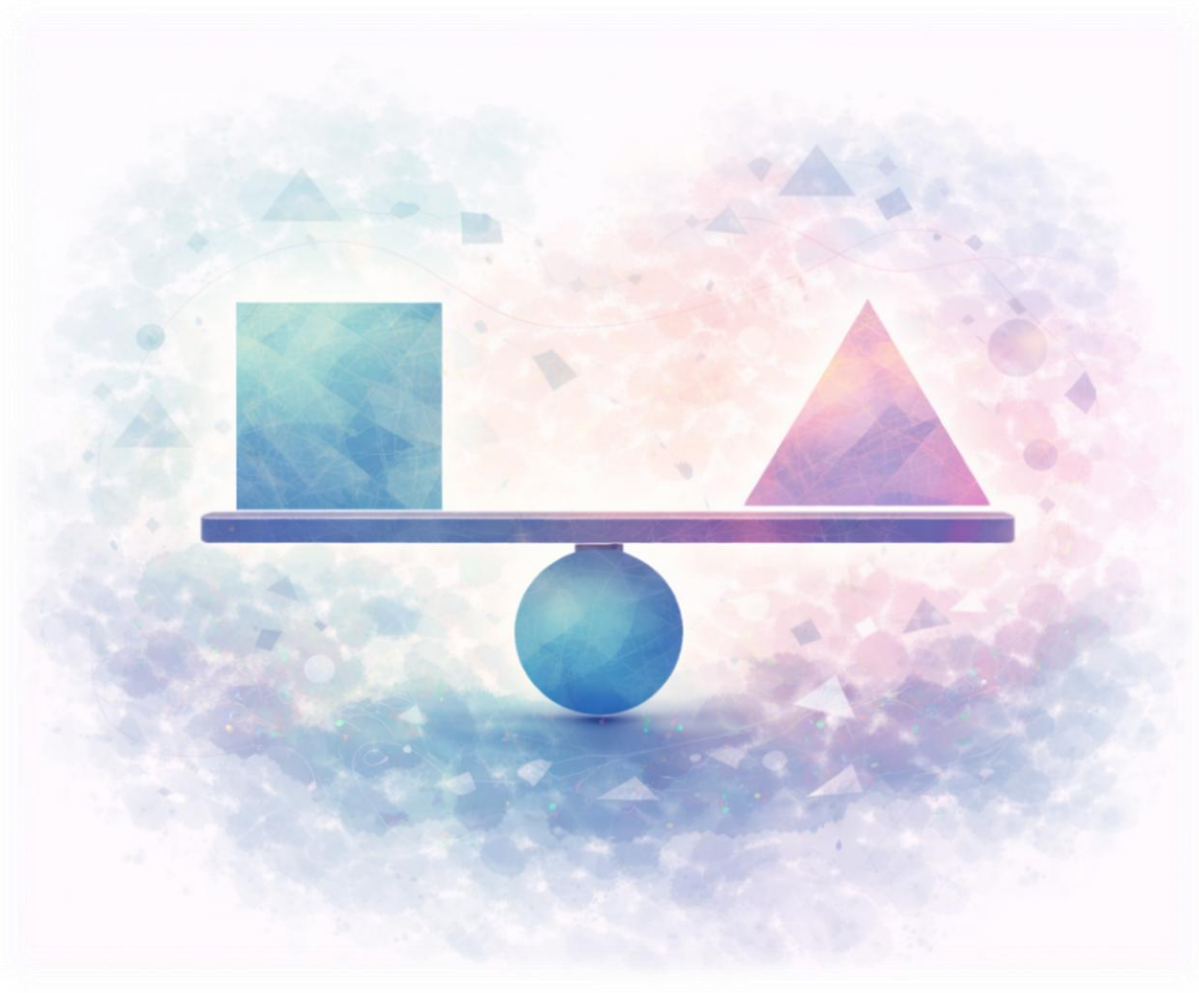
Earlier this week, we detected and responded to an intrusion into part of our production infrastructure. This one was different from anything we had handled before in one important way: it was driven, end to end, by an autonomous AI agent system - and we detected and dissected it largely with AI of our own.

external datasets and to several credentials used by our other any partner or customer data was affected, and We have found no evidence of tampering with software supply chain (container images and

Limitations: Misalignment

- **Objective misalignment:**
- LLMs may optimize for producing a plausible or helpful response rather than for scientific correctness or the user's actual underlying goal.
- **Specification and reward gaming:**
- When objectives are incomplete or poorly specified, models can satisfy the literal task while violating its intended purpose or methodological constraints.
- **Escalation in agentic systems:**
- When models can autonomously plan and act, minor misunderstandings of the goal can propagate across multiple steps and produce substantial downstream errors.
- **Practical implication:**
- Goals, constraints, intermediate decisions, and outputs should remain explicitly defined, monitored, and independently validated.

Ethical Aspects



- Scientific Integrity & Authorship
- Representation, Discrimination & Hatred
- Data Privacy & Confidentiality
- Environmental Impact

Ethical Aspects: Scientific Integrity & Authorship

- **Risk of unintentional plagiarism:**
- Generated formulations may closely resemble existing literature without explicit citation, even when this is not detectable by the user.
- **Blurred provenance of ideas:**
- LLM-assisted text can obscure whether arguments, phrasing, or structures originate from the user or from prior published work.
- **Disclosure and transparency:**
- Use of LLMs in scientific writing, analysis, or coding must usually be disclosed within journal, institutional, or funding guidelines.
- **Practical implication:**
- LLMs may support drafting and restructuring, but responsibility for content, originality, and proper citation remains entirely with the researcher.

Ethical Aspects: Scientific Integrity & Authorship

RESEARCH

Open Access

False authorship: an explorative case study around an AI-generated article published under my name



Diomidis Spinellis^{1,2*}

EDITORIAL ▶ eNeuro. 2026 Jan 7;13(1):ENEURO.0470-25.2025. doi: [10.1523/ENEURO.0470-25.2025](https://doi.org/10.1523/ENEURO.0470-25.2025)

The ethics of erroneous AI-generated scientific figures

Original Paper | [Open access](#) | Published: 14 June 2025

Volume 27, article number 31, (2025) [Cite this article](#)

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AI-Generated Scientific Papers: Crisis? What Crisis?

[Christophe Bernard](#)

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PMCID: PMC12810629 PMID: [41530046](#)

Research Article

Is AI my co-author? The ethics of using artificial intelligence in scientific publishing

[Barton Moffatt](#) & [Alicia Hall](#)

Pages 1313-1329 | Received 15 May 2024, Accepted 26 Jul 2024, Published online: 07 Aug 2024

[Cite this article](#) <https://doi.org/10.1080/08989621.2024.2386285>



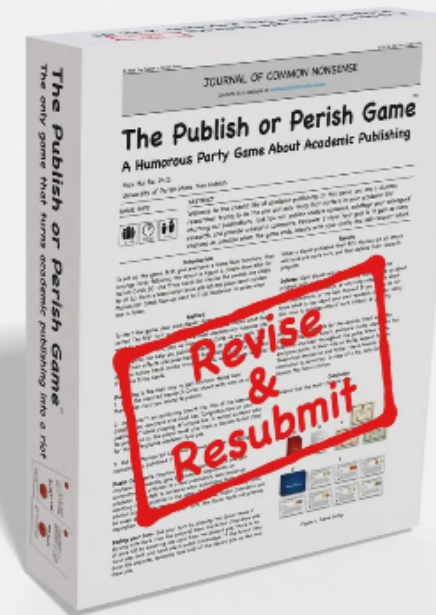
Ethical Aspects: Scientific Integrity & Authorship

A Game Where We Embrace Plagiarism and Academic Chaos

The Publish or Perish Game is a humorous party game about academic publishing. Players race to publish manuscripts with useless nonsense while sabotaging each other's research. Suitable for academics seeking catharsis and families who don't mind light emotional damage.

Ages: 14+ | **Players:** 3–6 | **Playtime:** 10 minutes per player

<https://publishorperish.games>



Ethical Aspects: Representation, Discrimination & Hatred

- **Bias in training data:**
- LLMs are trained on large-scale text corpora that reflect historical, cultural, and structural biases present in society and academia. Certain populations, regions, languages, disciplines, or perspectives may be systematically underrepresented or misrepresented.
- **Risk of biased outputs:**
- Generated text reinforces stereotypes, normative assumptions, or dominant viewpoints.
- **Scientific implications:**
- Biases can influence problem framing, interpretation of results, and prioritization of research questions.
- **Practical implication:**
- Outputs should be critically reviewed for fairness, inclusivity, and representational balance, especially in sensitive or policy-relevant contexts.

Ethical Aspects: Representation, Discrimination & Hatred

Review Article | [Open access](#) | Published: 13 September 2023

Ethics and discrimination in artificial intelligence-enabled recruitment practices

► *AI Ethics*. 2022 Feb 10;2(4):771–787. doi: [10.1007/s43681-022-00138-8](https://doi.org/10.1007/s43681-022-00138-8)

[Zhisheng Chen](#) 

Humanities and Social Sciences Communications 10, Article nu

AI bias: exploring discriminatory algorithmic decision-making models and the application of possible machine-centric solutions adapted from the pharmaceutical industry

[Lorenzo Belenguer](#)^{1,✉}

Letters

Facial recognition technology discriminates against people of colour

Alba Kapoor from Amnesty International responds to an article about the Met's increased use of the technology

France will investigate Musk's Grok after AI chatbot posted Holocaust denial claims

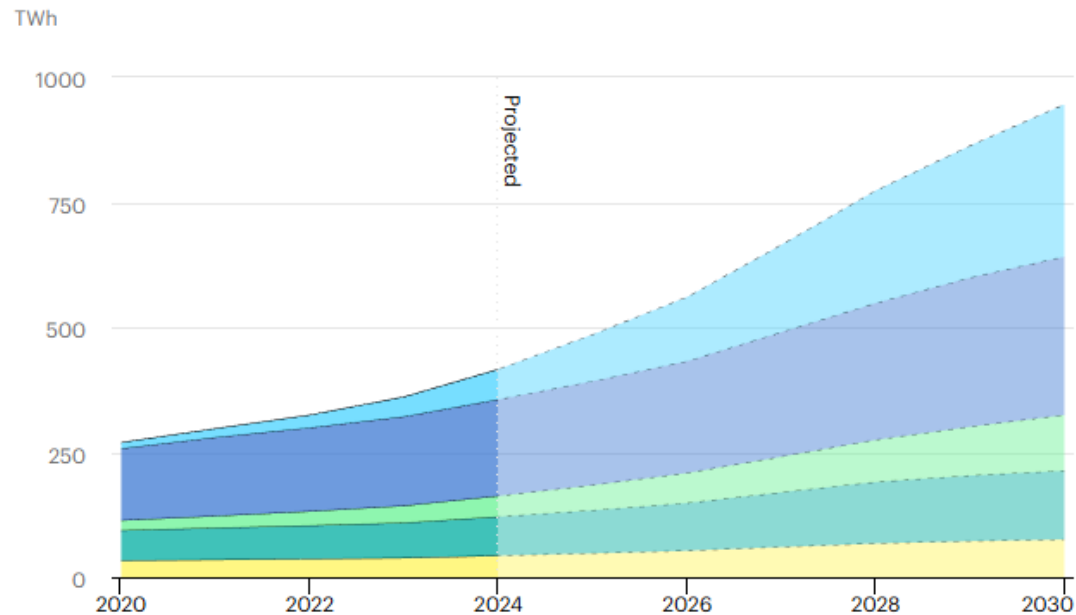
World Nov 21, 2025 5:47 PM EST

Ethical Aspects: Data Privacy & Confidentiality

- **No assumption of confidentiality:**
- Interactions with LLMs should not be assumed to be private, especially when using free or publicly hosted services as user inputs may be logged and, depending on service and settings, used for system monitoring or model improvement.
- **Risk of sensitive data exposure:**
- Entered information may include unpublished results, personal data, patient information, or proprietary content that should not be shared without safeguards.
- **Regulatory and legal obligations:**
- Use of LLMs must comply with applicable data protection regulations (e.g. GDPR), institutional policies, and ethical guidelines.
- **Practical implication:**
- Do not input confidential, identifiable, or regulated data unless explicit technical, legal, and contractual protections are in place.

Ethical Aspects: Environmental Impact

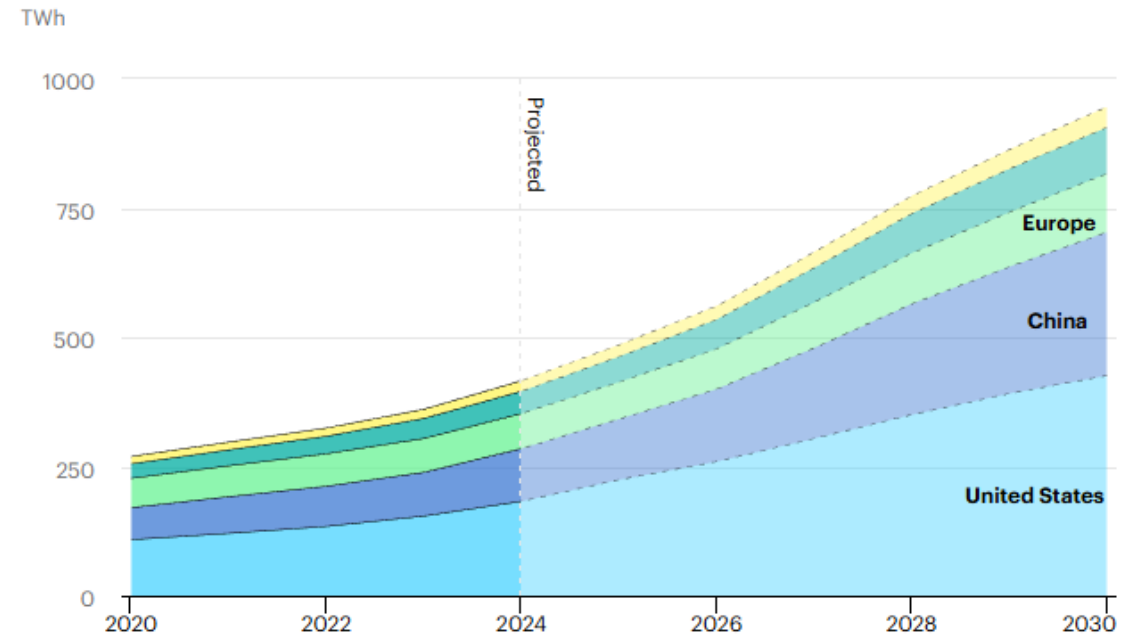
Global data centre electricity consumption, by equipment, Base Case, 2020-2030



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Accelerated servers Conventional servers Other IT equipment Cooling
Other infrastructure

Open Data centre electricity consumption by region, Base Case, 2020-2030



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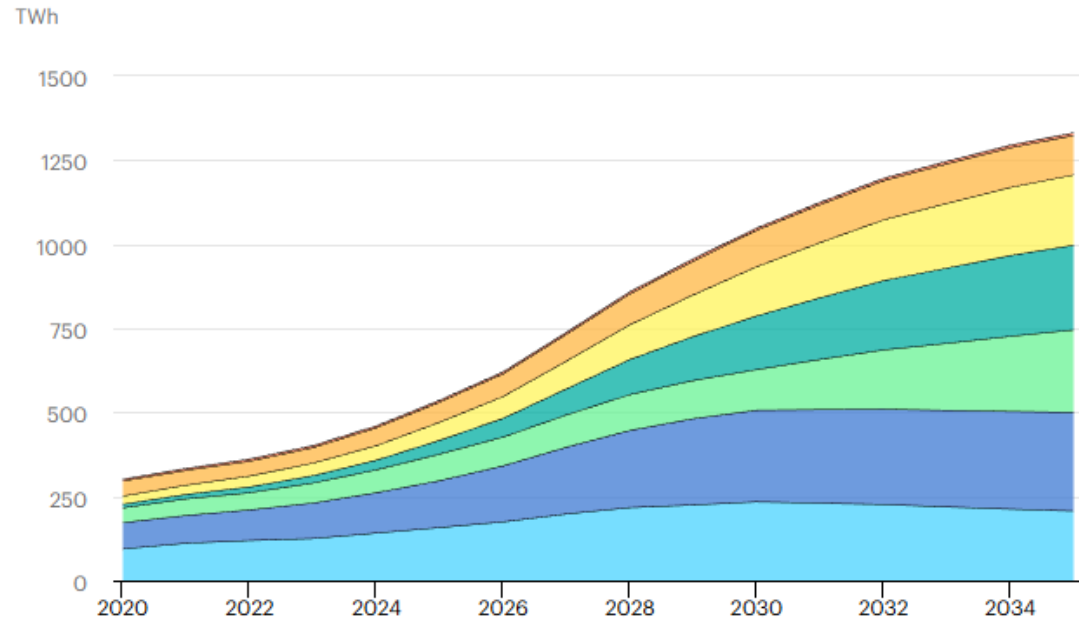
United States China Europe Asia excluding China Rest of world

<https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai>

Ethical Aspects: Environmental Impact

Sources of global electricity generation for data centres, Base Case, 2020-2035

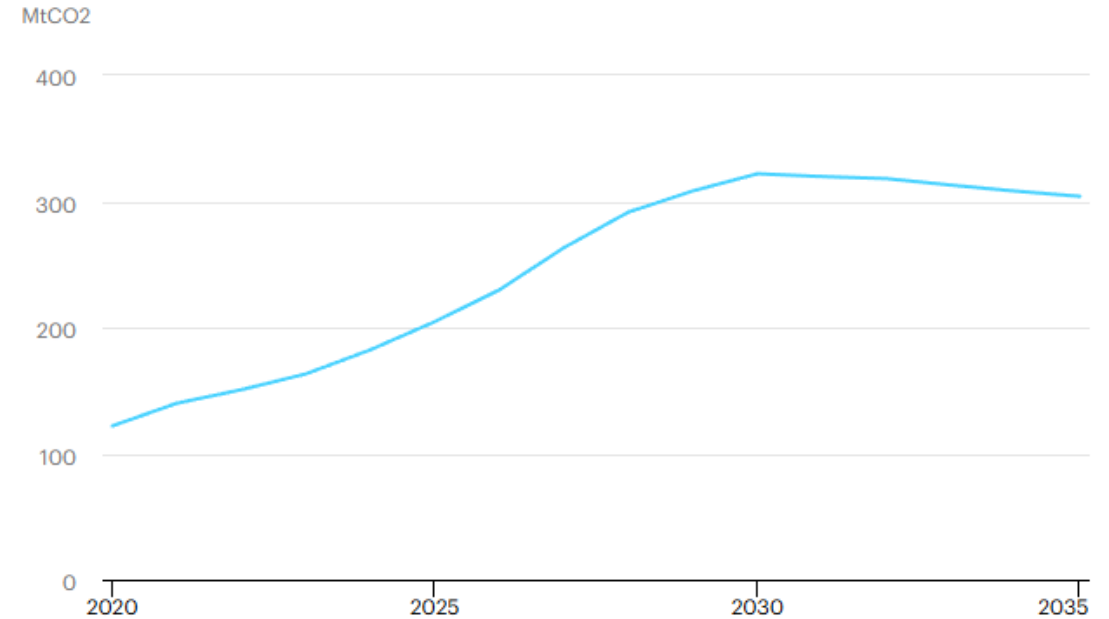
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Global data centre CO2 emissions, Base Case, 2020-2035

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<https://www.iea.org/reports/energy-and-ai/energy-supply-for-ai>



Thank you!

Your Questions are Appreciated!